Contents.

1 Introduction

At the most basic level, a natural language model is a function that takes as input a sequence of words and attempts to predict the word that follows. This prediction takes the form of a probability distribution over the set of all possible words, indicating the level of confidence the model has that any given word might follow. This model generates text on a given input by sampling from the output distribution. Once a word is generated, it is concatenated to the end of the original input, leading to a new input for the model to once again sample from. Repeating this process yields output text as desired.

More precisely, text is segmented into *tokens*. Although tokens are often words, they take on many other forms: punctuation, prefixes and postfixes, single characters. Ideally, these tokens would allow for a model that is not restricted to one language or type of text, and could make meaningful predications on any form of structured text with sufficient training data (other languages, code, etc.). We will assume that the set $\mathcal{X}$ of tokens for our model is chosen beforehand, but note that determining this set is an important part of engineering an efficient language model.

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A genius is the one most like himself.
- Thelonious Monk

Ma parole, c'est fait! 50 buts!

三人行，必有我师。

\frac{\bar{X}_n - \mu}{\sigma / \sqrt{n}}
\xrightarrow{d} \mathcal{N}(0,1)

while True:
    print("This will never end!")
```

Figure 1: GPT-4 tokenizer on sample text.

Let n be the number of tokens in the sequence of text the model takes in as input, known as *context length*. The model can therefore be seen as a function $f : \mathcal{X}^n \to \mathcal{P}_\mathcal{X}$, where $\mathcal{P}_\mathcal{X}$ is the set of probability distributions over $\mathcal{X}$. To develop a model, we design the model's parametric form $f(\boldsymbol{X} \mid \theta)$ and use likelihood estimation on a large set of sample data to determine parameters θ for which model performs sufficiently well. These are not simple tasks: for any decent model, f is extremely complex and highly nonlinear, and the size of $\mathcal{X}$, n, and θ are several orders of magnitude beyond the regime on which classical methods in statistics are effective.

Model	Tokens $\mathcal{X}$	Context n	Parameters θ	Release
BERT (Large)	30000	512	340 Million	2018
GPT-2	50257	1024	1.5 Billion	2019
GPT-3	50257	2048	175 Billion	2020
Llama 3	128000	128000	405 Billion	2024

Table 1: Size of $\mathcal{X}$, n, and θ of some modern language models.

Assuming that there exists methods to optimize parameters θ with respect to model performance, this approach to developing a model is inherently powerful. Models can first be designed using intuition about how a machine may interpret and predict language under a mathematical framework. If the model performs sufficiently well after training, the goal is accomplished. From a purely predictive perspective, it is not necessary that the learned internal mechanisms resemble the original intuition motivating the architechture.

Earlier approaches to language modeling relied on architectures (RNN, CNN) which primarily capture information in a limited or compressed way as text is processed sequentially. These models have difficulty maintaining a truly global understanding of a longer passage, since dependencies from distant parts of the text are either truncated or gradually lost as information is passed through successive steps.

Introduced in 2017, the Transformer architecture [Vas+17] resolves this issue by introducing a completely different mechanism, one which allows every token to directly interact with every other token in the input. This shift fundamentally changed the field of natural language processing, ultimately powering the large language models that now underpin modern AI systems. Section 2 gives an intuition-based overview of the Transformer architecture.

Among other ways to scale a model, of particular interest is the context length n. In theory, this would allow the model to use much longer-range context, enabling it to connect information across distant parts of a document and maintain a more coherent global understanding of the input. In real-world applications, this is especially useful for tasks like summarizing long reports, analyzing legal or financial documents, and answering questions that require information spread across many pages. It also improves performance in settings like code generation, where earlier definitions or functions must be remembered much later in the file.

However, scaling n has been shown to be a non-trivial task. Among other reasons, the attention mechanism core to the Transformer is suceptible to degenerate behaviour in the large-n limit, which leads to complete loss of expressive power of the model. Recent work has focused on improving the scaling behaviour of softmax attention by introducing an extra parameter, which can be found proposed in [Nak25]. Empirically, various modifications have been shown to mitigate the effects of this collapsing attention (REFERENCES), but mathematical justification has remained incomplete. The rest of this report summarizes relevant results in an attempt to unify existing empirical observations and theoretical perspectives on attention scaling.

2 The Transformer

This section provides a precise description of the architecture of the transformer model as outlined in [Vas+17]. For each step, a brief heuristic justification of the design choice is given based on intuition about how a machine might use the architecture to model language effectively, however it is important to keep in mind that all that is being done is designing the parametric form $f_\theta(\mathbf{X})$, from which the model will optimize θ over a set of training data. Whether the model “thinks” this way in practice is hard to determine, and likely not the case.

Throughout, any *weight matrix* W or *bias vector* b is a learned parameter of the model, and any other value is either an input or an intermediary value. In a computational spirit, we will use the "←" symbol to indicate variable assignment. Vector arrows will be used in order to clarify when individual rows or columns of matrices are in question.

2.1 Softmax

Beyond language modelling, it is frequently needed that a model output a probability distribution over a finite set of candidate outcomes. This arises in tasks such as classification, retrival, and decision-making, where the model must assign relative likelihoods to discrete alternatives rather than produce a single deterministic value. In these settings, the outputs are typically unconstrained real-valued scores that must be converted into a valid probability distribution.

Definition 2.1.1 (Softmax): Let $s \in \mathbb{R}^n$. Define the function $\mathsf{softmax} : \mathbb{R}^n \to \mathcal{P}_n$ as

$$\mathsf{softmax}(s)_j = \frac{e^{s_j}}{\sum_{k=1}^{n} e^{s_k}}.$$

For a matrix $S \in \mathbb{R}^{n \times m}$, $\mathsf{softmax}(S)$ is the $\mathbb{R}^{n \times m}$ matrix where $\mathsf{softmax}$ has been applied either row-wise or column-wise, depending on context.

It is clear that $\mathsf{softmax}$ outputs a valid probability distribution, is differentiable, and preserves the ordering of scores. While many other functions could in principle be used, $\mathsf{softmax}$ is the most widely adopted choice.

2.2 Embedding and Positional Encoding

We assign each token in $\mathcal{X}$ is assigned a unique number from 1 to $|\mathcal{X}|$ and represent it as one-hot vector[1]. Thus, our input $X = \left(\vec{X}_1, ..., \vec{X}_n\right) \in \mathcal{X}^n$ is represented as a $|\mathcal{X}| \times n$ matrix, where each column is a one-hot encoding of a token.

Modeling and learning are much more difficult in high-dimensional discrete spaces, as the tools of calculus are not available. Hence, we choose an *embedding dimension* d which allows for a continuous representation of token "meaning" in $\mathbb{R}^d$. One interpretation motivating this embedding space is the idea that orthogonal directions could encode different concepts or ideas, and that linear combinations of vectors along these directions could represent the meaning of a given token.

A matrix $W_E \in \mathbb{R}^{d \times |\mathcal{X}|}$ contains all token embeddings, where column i is the embedding for token i. For any token $\vec{X}_i$, multiplication by this matrix gives the corresponding column of W_E.

[1] A one-hot encoding is a vector that has a 1 in the position corresponding to the number it represents, and 0 everywhere else.

$$\underbrace{\begin{pmatrix} 0.12 & -0.33 & \dots & 0.07 \\ 0.54 & 0.21 & \dots & -0.11 \\ \vdots & \ddots & \ddots & \vdots \\ -0.08 & 0.44 & \dots & 0.19 \end{pmatrix}}_{W_E} \underbrace{\begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}}_{\vec{X}_i}$$

For the complete input X, the columns of the matrix multiplication $W_E X$ give the corresponding embeddings for each token. This process is equivalent to a lookup table.

We also allow for the model to obtain information of the position of a token in the sequence. This is given by the matrix W_P, where column i contains the encoding for position i. Historically, this has been implemented in different ways, including hard-coding this matrix to follow a sinusoidal pattern, as in the original Transformer architecture, or learning the positional embeddings directly during training. More recent approaches incorporate positional information within the attention mechanism itself through techniques such as relative or rotary positional embeddings. Regardless of the implementation, the goal is to provide the model with information about the ordering of tokens, which would otherwise be absent from the attention mechanism. We will assume this is a learned parameter matrix.

Combined with the embedding step, we compute

$$E \leftarrow W_E X + W_P,$$

which contains all token-level information that is independent of any interactions between tokens.

2.3 The Attention Layer

Given that words in any language rarely have a unique meaning independent of context, it is essential to allow tokens to “communicate” with each other. The attention layer modifies the positions of tokens in embedding space to better suit their meaning in context.

2.3.1 Single-Headed Attention

Let $W_K, W_Q \in \mathbb{R}^{d_h \times d}$, where d_h is known as the *head dimension*. Given the previously obtained token embeddings E, we compute

$$Q \leftarrow W_Q E, \qquad K \leftarrow W_K E,$$

called the *query matrix* and *key matrix*, respectively. Note that the columns of these matrices are transformations of embedding vectors: $\vec{Q}_i = W_Q \vec{E}_i$ and $\vec{K}_i = W_K \vec{E}_i$. The query $\vec{Q}_i$ represents what information the token is looking for from other tokens in order to refine its contextual meaning. The key $\vec{K}_i$ represents what kinds of information the token can provide to other tokens. Both these vectors are projections of their corresponding token embeddings into a shared lower-dimensional space $\mathbb{R}^{d_h}$, specifically used for relational matching.

To capture the extent to which the key $\vec{K}_i$ is relevant to query $\vec{Q}_j$, we compute the *attention score* $s_{ij} = \vec{K}_i \cdot \vec{Q}_j$, where larger values correspond to alignment in the query-key space. Thus, the matrix

$$S \leftarrow \frac{K^T Q}{\sqrt{d_h}}$$

has scaled attention score $s_{ij}/\sqrt{d_h}$ as entry (i, j). This scaling factor keeps dot-product similarity dimension-invariant, meaning scores stay probabilistically $O(1)$ scale regardless of head dimension and is derived assuming W_Q and W_K have entries with mean 0 and variance $1/d_h$, as is common in the initialization of models and often enforced through various normalization procedures beyond the scope of this overview.

Given a query, we would like these scores to correspond to weights indicating the relative importance of each key's token for updating the embedding of the queried token. We therefore apply the softmax function column-wise to S giving

$$A \leftarrow \text{softmax}(S).$$

The following is an example of the above step.

	The	**dog**	**sat**	**down**
The	1.1	−0.2	−0.8	−0.5
dog	−0.3	2.4	3.6	0.7
sat	−0.7	1.8	2.9	1.2
down	−0.4	0.6	1.3	2.1

Scores S

$\overset{\text{softmax}}{\mapsto}$

	The	**dog**	**sat**	**down**
The	0.61	0.04	0.01	0.04
dog	0.15	0.56	0.62	0.14
sat	0.10	0.31	0.30	0.24
down	0.14	0.09	0.06	0.58

Weights A

This completes the conversion of token embeddings into a normalized distribution over which tokens are most relevant to each other.

Returning to the original embeddings E, let $W_V \in \mathbb{R}^{d \times d}$. we compute the *value matrix*

$$V \leftarrow W_V E.$$

The columns $\vec{V}_i = W_V \vec{E}_i$ represent the information each token contributes when it is used by other tokens. Recalling that the columns $\vec{A}_i$ of the matrix A contain the attention weights assigned by token i, the update to a token's embedding is given by the weighted sum of the value vectors according to weights $\vec{A}_i$. In matrix form, this can be written as

$$\Delta E \leftarrow VA.$$

This update is then applied to E to obtain the final output of the attention layer,

$$E_{\text{out}} \leftarrow E + \Delta E.$$

2.3.2 Multi-Headed Attention

We wish to use multiple single-headed attention layers in parallel to allow different heads to specialize in different types of relationships and patterns within the input sequence. Let h be the number of heads in this multi-headed attention layer. We choose the head dimension d_h such that $d = hd_h$.

As before, for each head, we have matrices $W_Q^{(i)}, W_K^{(i)} \in \mathbb{R}^{d_h \times d}$ and compute

$$A^{(i)} \leftarrow \operatorname{softmax}\left(\frac{K^T Q}{\sqrt{d_h}}\right).$$

Allowing for each $W_V^{(i)}$ matrix to have size $d \times d$ would result in hd^2 parameters across all heads. In practice, we would like the parameter count to be comparable to W_Q and W_K, so we force W_V to have the low-rank structure

$$W_V^{(i)} = W_{V_l}^{(i)} W_{V_r}^{(i)},$$

where $W_{V_l}^{(i)} \in \mathbb{R}^{d \times d_h}$ and $W_{V_r}^{(i)} \in \mathbb{R}^{d_h \times d}$. Across all heads, this contributes $2dd_h$ parameters.

While this factored matrix could be applied per-head, each attention layer instead computes $W_{V_r}^{(i)} A$, which are then concatenated vertically and multiplied by the horizontal concatenation of $W_{V_l}^{(i)}$ matrices, giving[2]

$$\Delta E \leftarrow \underbrace{\begin{bmatrix} W_{V_l}^{(1)} & \cdots & W_{V_l}^{(h)} \end{bmatrix}}_{W_O} \begin{bmatrix} W_{V_r}^{(1)} A \\ \vdots \\ W_{V_r}^{(h)} A \end{bmatrix},$$

which is equivalent to computing $\Delta E^{(i)} \leftarrow W_V^{(i)} A$ for every head and summing the results. Because of this, we will consider the horizontal concatenation W_O to be a separate matrix that is applied after concatenating the results from each individual head. For each head, we will consider the $W_{V_r}^{(i)}$ matrix to simply be the $W_V^{(i)}$ matrix applied in the single-headed case.

Adding this to the original embeddings gives the final output for the multi-headed attention,

$$E_{\text{out}} \leftarrow E + \Delta E.$$

2.3.3 Masking

In self-attention, every token can attend to every other token. However, in many settings including language modelling, we wish to restrict which tokens are allowed to interact. In most generality, if we wish to forbid a token from interacting with another, we can set the corresponding entry in S to $-\infty$, which leads to a corresponding entry in A to be 0. In

[2]This allows the transformation to be expressed as a composition of two learned linear maps, which is computationally more efficient and better structured for parameter optimization than a product of two matrices.

practice, this is done by adding a masking matrix M with a very large negative integer in the desired entries and 0 elsewhere to E.

In the context of language modelling, this is espeically useful in order to parellelize training of th model. For example, the sequence "the dog sat down" contains four sub-examples:

1. [empty context] $\to$ the,
2. the $\to$ dog,
3. the dog $\to$ sat,
4. the dog sat $\to$ down.

An appropriate mask for this purpose needs to prohibit tokens from communicating with tokens beyond their current position. In other words, query i cannot access key j for any $j > i$, meaning that mask must be applied to the strictly lower triangular part.

	The	**dog**	**sat**	**down**
The	1.1	-0.2	-0.8	-0.5
dog	-∞	2.4	3.6	0.7
sat	-∞	$-\infty$	2.9	1.2
down	-∞	$-\infty$	$-\infty$	2.1

Masked Scores S

$\overset{\text{softmax}}{\mapsto}$

	The	**dog**	**sat**	**down**
The	1	0.07	0.01	0.04
dog	0	0.93	0.66	0.14
sat	0	0	0.33	0.24
down	0	0	0	0.58

Weights A

Although essential to include in this overview, the mathematical analysis in future sections does not apply masking.

2.4 Feed Forward Layer

Once a token has been contextualized using attention, it carries information from across the sequence, but this information is still only combined linearly through a weighted sum. The feed forward layer allows for each token to independently introduce nonlinear transformations in its embedding, converting this context into more expressive and higher-level representations. This is done using a *multi-layer perceptron*.

Let E be input embeddings that have been processed by an attention layer. The multi-layer perceptron aims to model how a biological brain might process information using neurons. We choose a feed-forward dimension d_f, which we will interpret to be the number of neurons in this layer.

Each neuron processes information using an affine transformation of the input data. This is then fed into a non-linear *activation function* σ which aims to model how a neuron might fire. We use $\sigma(x) = \max(0, x)$, known as ReLU, although there are many other options. The *activation* of neuron i for some embedding $\vec{E}_j$ is the scalar value given by

$$n_{ij} \leftarrow \sigma\left(\overrightarrow{W}_{\text{in}}^{(i)} \cdot \vec{E}_j + b_{\text{in}}^{(i)}\right).$$

Letting W_{in} to be the matrix with weights $\overrightarrow{W}_{\text{in}}^{(j)}$ as columns and b_{in} to be the column vector with components $b_{\text{in}}^{(i)}$, this can be expressed in matrix form as

$$N \leftarrow \sigma\Big(W_{\text{in}}E + b_{\text{in}}\mathbf{1}_{d_f}^T\Big),$$

where σ is applied component-wise, and $\mathbf{1}_{d_f} \in \mathbb{R}^{d_f}$ is the column vector of all ones.

Similar to attention, an update to each embedding is calculated using an affine transformation of the embeding's corresponding neuron activations. In parallel, this gives

$$\Delta E \leftarrow W_{\text{out}}N + b_{\text{out}},$$

which is added to the input to give

$$E_{\text{out}} = E + \Delta E.$$

2.5 Unembedding

In practice, embeddings are fed through many iterations of transformer blocks (a multi-headed attention layer followed by a feed forward layer), each with separate parameters. After this processing, we wish to convert each embedding $\vec{E}_i$ into a probability distribution over tokens $\mathcal{X}$, indicating the level of confidence that each token in the vocabulary could be the next token at position i.

This is done by applying the unembedding matrix $W_U \in \mathbb{R}^{|\mathcal{X}| \times d}$ to $\vec{E}_i$, followed by a softmax. In matrix form,

$$Y = \text{softmax}(W_U E + b_U \mathbf{1}^T),$$

where softmax is applied column-wise. Note that each column $\vec{Y}_i$ is the prediction for the next token at each position i in the input sequence. If lower triangular masking is applied, this allows for n different predictions on the n different sub-examples. This is useful for model training purposes, but for generation purposes we only use the distribution $\vec{Y}_n$ corresponding to the last input token.

2.6 Summary

The following figure summarizes the attention mechanism. Terminology varies slightly.

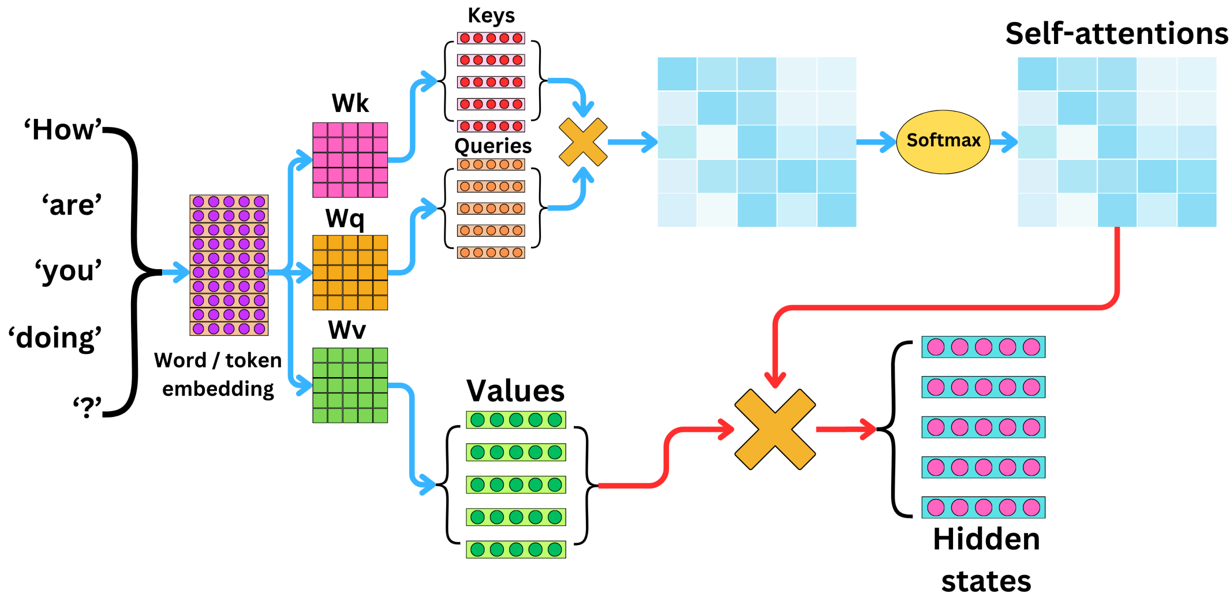

Figure 2: The Attention Mechanism. Source: The AI Edge Newsletter [Ben24]

2.7 Long-Context Transformers

While the term "long" continuously changes meaning as advances are made in machine learning, long-context transformers are transformers that are designed to support a context length n that is substantially longer than what is seen in common models. Many real-world tasks such as reasoning over long documents, codebases, or dialogues require access to information that may be introduced thousands or even millions of tokens earlier, where simply truncating the context leads to severly degraded performance. Increasing context length effectively augments the models "memory", which allows for richer retrieval of relevant evidence and more coherent global reasoning.

It is important to distinguish between two types of long-context transformers. The first consists of models explicitly trained on long sequences, which attempt to internalize long-range dependencies during optimization. The second consists of models trained on shorter sequences but designed to generalize at inference time to longer sequences. There also exists hybrids of these two models, which might split training into 80% short sequences and 20% long sequences, for example.

From a scaling perspective, there are many issues related to naively increasing context length that apply to both types of long-context transformers. First and foremost, attention grows quadratically in context length, since a vanilla attention matrix A is of size $n \times n$. Second, attention suffers from large-n scaling issues inherent to the softmax function. The remainder of this report investigates the latter. In general, the results collected throught are moreso properties of the softmax function under specific inputs rather than anything regarding the underlying Transformer architechture or training, and thus we will not differentiate between the two types of long-context Transformers unless it is pertinent.

3 The Need for Attention Scaling

3.1 Insufficiency of Standard Softmax

3.1.1 Single Head

Consider the standard softmax function applied to a sequence of score vectors $\left(s^{(n)}\right)$ such that $s^{(n)} \in \mathbb{R}^n$ for all n with uniformly bounded components $m \leq s_j^{(n)} \leq M$ for $m, M \in \mathbb{R}$. For a fixed j,

$$\operatorname{softmax}\left(s^{(n)}\right)_j = \frac{e^{s_j}}{\sum_{k=1}^{n} e^{s_k}} \leq \frac{e^M}{ne^m} = \frac{e^{M-m}}{n},$$

as well as

$$\operatorname{softmax}\left(s^{(n)}\right)_j \geq \frac{e^{m-M}}{n}.$$

Then, we have that $\operatorname{softmax}\left(s^{(n)}\right)_j = \Theta\left(\frac{1}{n}\right)$ for all j and thus the maximum attention weight $a_{\max}$ goes to 0 as n goes to ∞. Heuristically, the entire softmax vector becomes diffuse and "true" attention, in the sense of assigning a non-vanishing proportion of mass to a distringuished set of coordinates, cannot emerge. Therefore, any nontrivial attention mechanism in the large-n limit must be driven by score fluctuations whose scale grows with n, allowing certain coordinates to overcome the $\Theta(n)$ growth of the softmax normalizing constant. Because attentions scores are usually designed to be probabilistically $O(1)$, this is naturally a problem when considering large-n inputs.

As an example, consider the sequence of vectors $s^{(n)} = (3, -2, ..., -2)$ with first component 3 and all other components -2, noting that these are clearly uniformly bounded. Figure 3 shows that not only does $a_{\max}$ decrease to 0 relatively quickly, but the entropy of the distribution quickly approaches the entropy of the uniform distribution. This is refered to as *rank collapse*, since it is the behaviour obtained when a rank one matrix is used in the attention calculation.

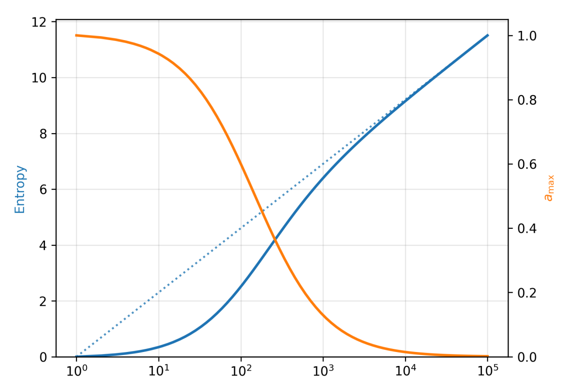

Figure 3: Rank collapse on scores with bounded components.

Even if scores do grow with n, the nature of the growth of the scores can lead to another failure mode known as *entropy collapse*. This occurs when the maximum component grows too quickly with respect to the rest of the scores, leading softmax to behave as a "hardmax" for large n, outputting a one-hot vector. This collapse is shown in Figure 4 for the sequence $t^{(n)} = \frac{1}{3}\log(n)s^{(n)}$.

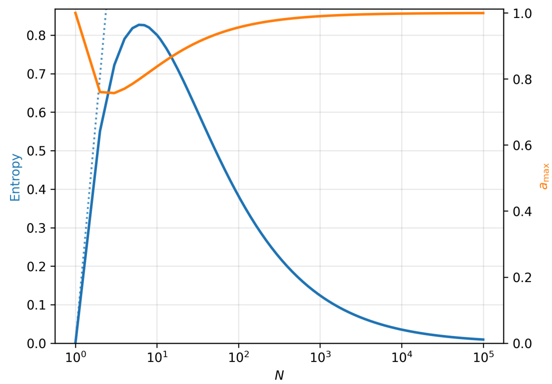

Figure 4: Entropy collapse on scores with growing components.

3.1.2 Transformer Setting

Theorem 3.1.1 (Long-Context Attention Collapse):

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a finite set of token embeddings, and let $X^{(n)} \in \mathcal{X}^n$ be a matrix of n embeddings.

Define query and key matrices $Q^{(n)} = \varphi\left(X^{(n)}\right) \in \mathbb{R}^{n\times k}$, $\quad K^{(n)} = \kappa\left(X^{(n)}\right) \in \mathbb{R}^{n\times k}$, where φ, κ are continuous functions obtained as compositions of a finite number L of layers of the following two types:

1. A feedforward layer $F_l(Z)_i = f_l(Z_i)$ that is acts row-wise and is continuous,
2. Self-attention layer $G_l(Z)_i = \sum_{j=1}^{n} a_{ij} v_l\left(Z_j\right)$, where $a_{ij} \in [0,1]$ are softmax-normalised attention coefficients and v is a continuous feedforward network.

Define the score matrix $S^{(n)} = Q^{(n)}\left(K^{(n)}\right)^T \in \mathbb{R}^{n\times n}$ and row-wise softmax attention matrix $A^{(n)} = \operatorname{softmax}\left(S^{(n)}\right)$.

Then, $\|A^{(n)}\|_{\max} \to 0$ as $n \to \infty$, where $\|A^{(n)}\|_{\max} = \max_{ij} |a_{ij}|$. [Vel+24]

Proof. Since $\mathcal{X} \subseteq \mathbb{R}^d$ is finite, both $\mathcal{X}$ and its convex hull $\operatorname{conv}(\mathcal{X})$ are compact. Since all feedfoward operators F_l and v_l act row-wise and are continuous, they preserve compactness row-wise. Each output row of a self-attention layer is a convex combination of vectors in a compact set, hence remains in a compact set. By induction over L, the finite number of layers, all embeddings lie in a compact set, independent of n. In particular, there exists a compact set $C \subseteq \mathbb{R}^k$ such that all rows $Q_i^{(n)}, K_i^{(n)} \in C$. Hence, there exists an $M > 0$ such that any score has the bound $|S_{ij}^{(n)}| \leq M$. By a similar derivation as before, we obtain

$$\frac{e^{-2M}}{n} \leq A_{ij}^{(n)} \leq \frac{e^{2M}}{n}.$$

Since M does not depend on n, $\|A^{(n)}\|_{\max} \to 0$ as $n \to \infty$. $\square$

While a simplified model, this shows a different fundamental limitation of Transformers. For a fixed trained model, there is no architectural mechanism that guarantees stable, non-degenerate attention behaviour as the sequence length n increases beyond the range seen during training. Although one might hope that the model could compensate by implicitly scaling scores with sequence length, the architecture does not provide a reliable way to represent or extrapolate n. Consequently, any implicit learned rescaling of W_K or W_Q matrices remains uniformly bounded, and cannot in general counteract the linear growth of the softmax normalization as $n \to \infty$.

3.2 Scalable Softmax

An initial instict may be to introduce a fixed parameter $\beta > 0$ to softmax,

$$\operatorname{softmax}(s)_j = \frac{e^{\beta s_j}}{\sum_{k=1}^n e^{\beta s_k}},$$

which suffers from the same issue as standard softmax when scores are bounded, since

$$\frac{e^{\beta(m-M)}}{n} \leq \operatorname{softmax}(s)_j \leq \frac{e^{\beta(M-m)}}{n}.$$

Hence, any non-degenerate scaling β must depend on n, i.e. $\beta = \beta_n$. The *scalable softmax* function

$$\operatorname{ssmax}(z)_j = \frac{e^{(sp_n+b)z_j}}{\sum_{k=1}^n e^{(sp_n+b)z_k}}$$

is proposed in [Nak25], where s and b are scalar learned parameters that are unique to each attention layer and head, and p_n is a learnable parameter shared across all layers and heads, depending only on the input vector size n. They train this scalable softmax on a standard model and find a relationship fitting $p_n \sim \log n$. For a given attention layer and head (i.e. fixed constants), this corresponds to $\beta_n \sim \log n$ as proposed in our original setup.

According to these results, consider the scaling $\beta_n = \gamma \log n$, for some $\gamma > 0$, and assume that s has a unique maximum $z_{\max} > z_{\text{2nd}}$. On one hand, we have

$$a_{\max} = \frac{n^{\gamma z_{\max}}}{\sum_{k=1}^n n^{\gamma z_k}} \leq \frac{n^{\gamma z_{\max}}}{(n-1)n^{\gamma z_{\min}} + n^{\gamma z_{\max}}} = \frac{1}{(n-1)n^{-\gamma(z_{\max}-z_{\min})}+1}.$$

On the other hand,

$$a_{\max} \geq \frac{n^{\gamma z_{\max}}}{(n-1)n^{\gamma z_{\text{2nd}}} + n^{\gamma z_{\max}}} = \frac{1}{(n-1)n^{-\gamma(z_{\max}-z_{\text{2nd}})}+1}.$$

If $\gamma > \frac{1}{z_{\max}-z_{\min}}$, we see that $a_{\max} \to 1$, whereas if $\gamma < \frac{1}{z_{\max}-z_{\text{2nd}}}$, then $a_{\max} \to 0$. This is one example showing that even if β_n is of correct asymptotic order, constant multiples can still induce degenerate behaviour. Thus, β_n is likely quite fragile in a general sense, which motivates the study of attention scaling beyond the "quick fix" of adding a learnable parameter to the model.

Furthermore, we observe a phase transition phenomenon in γ. This phenomenon occurs beyond this simple example and appears to be intrinsic to the attention mechanism, as it appears in various different models of attention, as seen in later sections. Precisely, this phase transition separates the behaviour of attention into *subcritical* and *supercritical* regimes, $\gamma < \gamma_1$ and $\gamma > \gamma_2$ respectively for constants $\gamma_1 < \gamma_2$, in which the behaviour of attention is undesirable. For the *critical* regime $\gamma \in [\gamma_1, \gamma_2]$, often a single value $\gamma_c = \gamma_1 = \gamma_2$, attention behaves as desired. In terms of manually tuning this scaling, we see that this must proceed in two stages: determining the order of the scaling β_n followed by tuning the constant multiple γ.

Returning to the example $s^{(n)} = (3, -2, -2, ..., -2)$, we apply the scaling $\beta_n = 0.2 \log n$. As seen in figure Figure 5, this scaling avoids rank and entropy collapse.

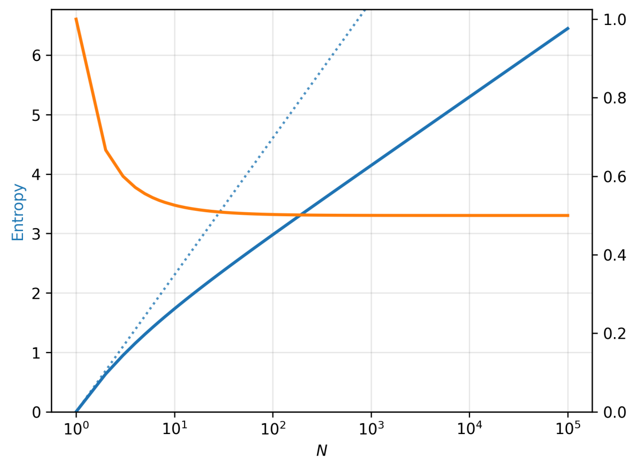

Figure 5: Non-degenerate behaviour of scaled softmax on $s^{(n)}$.

This scaling was obtained by the heuristic justification that to avoid both rank and entropy collapse, the maximum element $s_{\max}$ and the sum of all other elements must be of the same exponential order. This reasoning extends beyond this simple case to more complex score vectors, and is a first step in ensuring proper scaling. In this case,

$$a_{\max} = \frac{e^{3\beta}}{\underbrace{e^{3\beta}}_{\max} + \underbrace{(n-1)e^{-2\beta}}_{\text{bulk}}} = \left(1 + e^{-5\beta + \log n} + o\left(e^{-5\beta+\log n}\right)\right)^{-1},$$

which avoids convergence to either 0 or 1 (entropy and rank collapse, respectively) if $\beta = 0.2 \log n$, which correspond to entropy and rank collapse, respectively.

At this point, the question remains how to determine an effective scaling β_n in more general contexts, and whether any universality properties hold. Since transformer inputs lie between deterministic and fully random regimes, a completely general analysis seems intractable at the moment, and it is natural to first study simplified models that capture the essential structure of the attention mechanism before addressing more realistic settings.

4 Simplex-Like Geometry

The following is an overview of results and proofs found in [Che+25] with intermediate steps filled in as well as further discussion.

4.1 Setup

Consider a simplified version of unmasked single-headed attention in which the head dimension equals the embedding dimension and $W_K = W_Q = W_V = I_d$. For a collection of token embeddings $x_1, ..., x_n \in \mathbb{R}^d$, define the normalized vectors $u_i = x_i/\|x_i\|$. The scaled attention scores for the normalized vectors u_i are thus given by

$$s_{ij} = \beta\langle u_i, u_j\rangle,$$

where $\langle\cdot,\cdot\rangle$ denotes the standard inner product on $\mathbb{R}^d$, and β is a scaling factor to be chosen. The corresponding attention matrix A has entries

$$a_{ij} = \frac{e^{s_{ij}}}{Z_i}, \qquad Z_i = \sum_{k=1}^{n} e^{s_{ik}}.$$

The main object at study is the nonlinear operator $\mathrm{ATT} : \mathbb{R}^d \to \mathbb{R}^d$, defined as

$$\mathrm{ATT}(u_i) = \sum_{j=1}^{n} a_{ij} u_j.$$

Finally, recalling that in transformer architechtures, the attention layer computes an update to the original embeddings through a residual connection, we define the updated embeddings and their normalized versions

$$x_i' = \mathrm{ATT}(u_i) + \alpha x_i, \qquad u_i' = \frac{x_i'}{\|x_i'\|},$$

where $\alpha \geq 0$ is the strength of the residual connection. Geometrically, α interpolates between purely attention-driven dynamics and identity-preserving behaviour.

4.2 Critical Scaling

We are interested in how the attention scaling β affects the geometry of the token embeddings, namely how pairwise angles evolve under the attention update. Equivalently, this amounts to studying the relationship between $\langle u_i', u_j'\rangle$ and $\langle u_i, u_j\rangle$.

Consider a further simplified setup in which $\langle u_i, u_j\rangle = p \in (0, 1)$ if $i \neq j$ and $\|u_i\|^2 = q$. The key insight is that the behaviour of attention is governed by the normalization factor Z_i, which simplifies to

$$Z_i = e^{\beta} + (n-1)e^{p\beta}$$

under these assumptions. Thus, the asymptotic behaviour of Z_i depends on the competition between the self-attention term and the aggregate contribution of attention from all other tokens. Under the scaling $\beta = \gamma \log n$,

$$e^{\beta} = n^{\gamma}, \qquad (n-1)e^{p\beta} \asymp n^{1+p\gamma}.$$

Hence, the dominant contribution to Z_i depends on the comparison between the exponents γ and $1+p\gamma$. The critical threshold occurs when $\gamma = \frac{1}{1-p}$. This reveals that there are three regimes under which attention operates. In the subcritical regime $\gamma < \frac{1}{1-p}$, attention weights become asymptotically uniform, so each token moves toward the global average. In the supercritical regime $\gamma > \frac{1}{1-p}$, the self-attention term dominates, and attention operator converges to the identity map. In the critical regime $\gamma = \frac{1}{1-p}$, both terms are balanced, leading to nontrivial dynamics as desired for an effective attention layer.

In Theorem 4.2.1 we generalize this setup beyond the symmetric simplex setting by allowing pairwise inner products to vary within fixed bounds. We show that the attention operator acts asymptotically as a contraction in the subcritical regime and as an identity in the supercritial regime. Corollary 4.2.6 provides exact results for the limiting inner product for the simplex case. Although [Che+25] claims that these results demonstrate that the scaling $\beta = \Theta(\log n)$ is intrinsic to the behaviour of attention mechanisms rather than an artifact of the simplex geometry, we will later see that this in fact not the case.

Theorem 4.2.1 (Phase Transition in Attention Geometry):

Assume there exist constants $q_1, q_2 > 0$ and $p_1, p_2 \in (0,1)$ such that $q_1 \leq \|x_i\|^2 \leq q_2$ and $p_1 \leq \langle u_i, u_j \rangle \leq p_2$ for any $i \neq j$, with $p_1 = \langle u_i, u_j \rangle$ for some i, j. Let $\beta = \gamma \log n$ with $\gamma > 0$.

1. If $\gamma < \frac{1}{1-p_1}$ then there is a constant $\varepsilon > 0$ depending on α, p_2, q_1, q_2 such that

$$\liminf_{n\to\infty} \min_{i\neq j} \langle u_i', u_j' \rangle \geq p_1 + \varepsilon.$$

2. If $\gamma > \frac{1}{1-p_2}$, then for any i,

$$\lim_{n\to\infty} \langle u_i', u_j' \rangle = \langle u_i, u_j \rangle.$$

[Che+25]

In the subcritical regime, the value of ε could in theory be very small, and thus the theorem should be interpreted as a qualitative phase transition rather than a quantitative improvement guarantee in this regime. In the supercritical regime, we obtain a much stronger result. Although the randomly sampled vectors found in Figure 6 do not necessarily satisfy the assumptions of Theorem 4.2.1 and the embedding dimension $d = 2$ might lead to weaker concentration results, we still observe a clear separation between the subcritical (left) and supercritical (right) regimes, consistent with the qualitative behaviour described by the theorem.

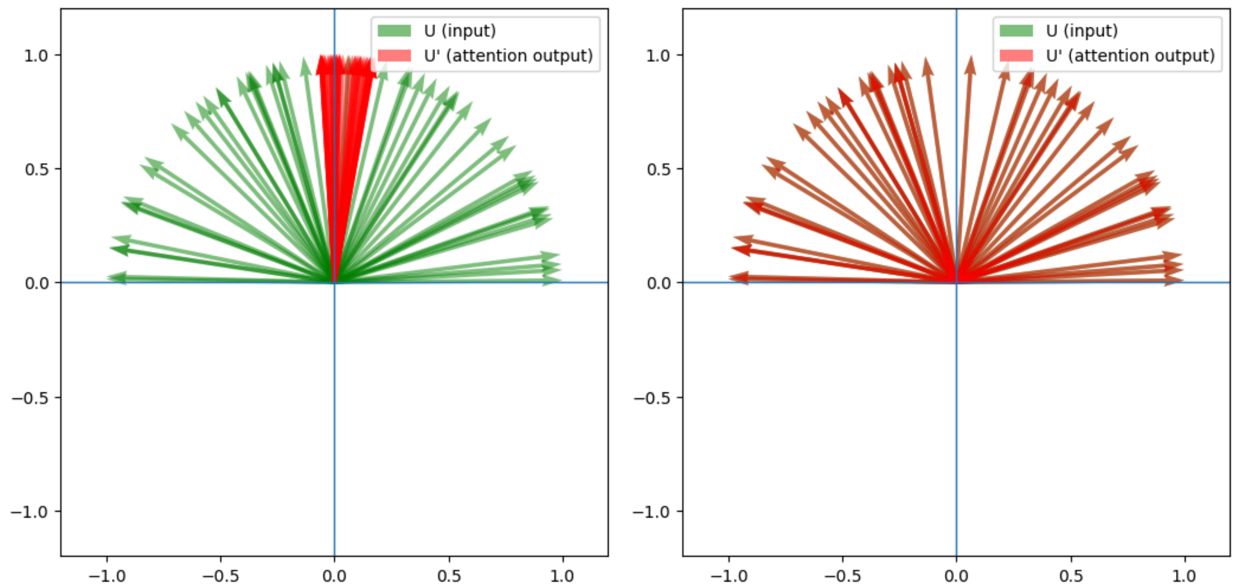

Figure 6: Behaviour of geometric attention on random vectors.

We will first prove a series of lemmas before proving Theorem 4.2.1.

Lemma 4.2.2: For any $i \in \{1, ..., n\}$, we have

$$Z_i = \begin{cases} (1 + o_n(1)) \cdot \left(\sum_{k \neq i} e^{s_{ik}}\right) & \gamma < \frac{1}{1-p_1}, \\ (1 + o_n(1)) \cdot e^{\beta} & \gamma > \frac{1}{1-p_2}. \end{cases}$$

Proof. Recall $s_{ik} = \beta \langle u_i, u_k \rangle$ and $\beta = \gamma \log n$, giving

$$Z_i = \sum_{k=1}^{n} e^{s_{ik}} = n^{\gamma} + \sum_{k \neq i} n^{\gamma \langle u_i, u_j \rangle}$$

If $\gamma < \frac{1}{1-p_1}$, then $\gamma - \gamma p_1 - 1 < 0$. Using $\langle u_i, u_j \rangle \geq p_1$,

$$\frac{n^{\gamma}}{\sum_{k \neq i} n^{\gamma \langle u_i, u_j \rangle}} \leq \frac{n^{\gamma}}{(n-1) n^{\gamma p_1}} = n^{\gamma - \gamma p_1 - 1} \cdot \frac{1}{1 - 1/n}$$

which goes to 0 as $n \to \infty$. Hence,

$$Z_i = \left(1 + \frac{n^{\gamma}}{\sum_{k \neq i} n^{\gamma \langle u_i, u_j \rangle}}\right) \left(\sum_{k \neq i} n^{\gamma \langle u_i, u_j \rangle}\right) = (1 + o_n(1)) \left(\sum_{k \neq i} e^{s_{ik}}\right).$$

Similarly, if $\gamma > \frac{1}{1-p_2}$, then $\gamma p_2 + 1 - \gamma < 0$. Using $\langle u_i, u_j \rangle \leq p_2$,

$$\frac{\sum_{k \neq i} n^{\gamma \langle u_i, u_j \rangle}}{n^{\gamma}} \leq \frac{(n-1) n^{\gamma p_2}}{n^{\gamma}} = n^{\gamma p_2 + 1 - \gamma}(1 - 1/n)$$

which goes to 0 as $n \to \infty$. Hence,

$$Z_i = \left(1 + \frac{\sum_{k \neq i} n^{\gamma \langle u_i, u_j \rangle}}{n^\gamma}\right)(n^\gamma) = (1 + o_n(1)) \cdot e^\beta.$$

$\square$

Lemma 4.2.3: If $\gamma > \frac{1}{1-p_2}$, then for any $i = \{1, ..., n\}$,

$$\mathrm{ATT}(u_i) = u_i + o_n(1),$$

where $\boldsymbol{o}_n(1)$ does not depend on i.

Proof. By the previous lemma,

$$\mathrm{ATT}(u_i) = Z_i^{-1}\left(e^\beta u_i + \sum_{j \neq i} e^{s_{ij}} u_j\right) = (1 + o_n(1))\left(u_i + e^{-\beta} \sum_{j \neq i} e^{s_{ij}} u_j\right),$$

using the fact that $(1 + o_n(1))^{-1} = 1 + o_n(1)$. Since $\|u_j\| = 1$ for all j,

$$\left\|e^{-\beta} \sum_{j \neq i} e^{s_{ij}} u_j\right\| \leq e^{-\beta} \sum_{j \neq i} e^{s_{ij}} \leq n^{-\gamma} \cdot n^{\gamma p_2}(n-1)$$

which goes to 0 as $n \to \infty$, independent of i. $\square$

Lemma 4.2.4: For any $i \in \{1, ..., n\}$,

1. If $\gamma < \frac{1}{1-p_1}$,

$$\|x_i'\|^2 \leq \alpha^2 \|x_i\|^2 + 2\alpha \|x_i\| p_2 + p_2 + o_n(1).$$

2. If $\gamma > \frac{1}{1-p_2}$,

$$\|x_i'\|^2 = (\alpha \|x_i\| + 1)^2 + o_n(1).$$

In both cases, $o_n(1)$ does not depend on i.

Proof. Using the definition of the update rule, expanding the inner product gives

$$\|x_i'\|^2 = \alpha^2 \|x_i\|^2 + 2\alpha \|x_i\| \langle u_i, \mathrm{ATT}(u_i) \rangle + \|\mathrm{ATT}(u_i)\|^2.$$

For the case of $\gamma < \frac{1}{1-p_1}$, Lemma 3.1 gave $Z_i = (1 + o_n(1))\left(\sum_{j \neq i} e^{s_{ij}}\right)$. Hence,

$$\begin{aligned}\langle u_i, \mathrm{ATT}(u_i)\rangle &= \langle u_i, \frac{1}{Z_i}\sum_{j=1}^{n} e^{s_{ij}} u_j\rangle \\ &= \frac{1}{Z_i}\sum_{j=1}^{n} e^{s_{ij}}\langle u_i, u_j\rangle \\ &\le \frac{1}{Z_i}\left(e^{\beta} + p_2\sum_{j\neq i} e^{s_{ij}}\right) \\ &= p_2 + o_n(1),\end{aligned}$$

since $e^{\beta}/\sum_{j\neq i} e^{s_{ij}} = o_n(1)$, as shown previously. Similarly,

$$\begin{aligned}\|\mathrm{ATT}(u_i)\|^2 &= \langle\frac{1}{Z_i}\sum_{k=1}^{n} e^{s_{ik}} u_k, \frac{1}{Z_i}\sum_{l=1}^{n} e^{s_{il}} u_l\rangle \\ &= \frac{1}{Z_i^2}\sum_{k=1}^{n}\sum_{l=1}^{n} e^{s_{ik}+s_{il}}\langle u_k, u_l\rangle \\ &\le \frac{1}{Z_i^2}\left(e^{2\beta} + 2p_2 e^{\beta}\sum_{j\neq i} e^{s_{ij}} + p_2\sum_{k\neq i}\sum_{l\neq i} e^{s_{ik}+s_{il}}\right) \\ &= p_2 + o_n(1).\end{aligned}$$

For the case of $\gamma > \frac{1}{1-p_2}$, applying Lemma 4.2.3 directly gives

$$\begin{aligned}\|x_i'\|^2 &= \alpha^2\|x_i\|^2 + 2\alpha\|x_i\|\langle u_i, u_i + \boldsymbol{o}_n(1)\rangle + \langle u_i + \boldsymbol{o}_n(1), u_i + \boldsymbol{o}_n(1)\rangle \\ &= \alpha^2\|x_i\|^2 + 2\alpha\|x_i\| + 1 + o_n(1) \\ &= (\alpha\|x_i\| + 1)^2 + o_n(1).\end{aligned}$$

□

Lemma 4.2.5: For $i, j \in \{1, ..., n\}$ with $i \neq j$,

1. If $\gamma < \frac{1}{1-p_1}$,

$$\langle x_i', x_j'\rangle \ge p_1(\alpha\|x_i\| + 1)\big(\alpha\|x_j\| + 1\big) + o_n(1).$$

2. If $\gamma > \frac{1}{1-p_2}$,

$$\langle x_i', x_j'\rangle = (\alpha\|x_i\| + 1)\big(\alpha\|x_j\| + 1\big)\langle u_i, u_j\rangle + o_n(1).$$

In both cases, $o_n(1)$ does not depend on i.

Proof. Expand the inner product to give

$$\begin{aligned}\langle x_i', x_j'\rangle = {}& \alpha^2\|x_i\|\|x_j\|\langle u_i, u_j\rangle + \alpha\|x_i\|\langle u_i, \mathrm{ATT}\big(u_j\big)\rangle \\ &+ \ \alpha\|x_j\|\langle u_j, \mathrm{ATT}(u_i)\rangle + \langle\mathrm{ATT}(u_i), \mathrm{ATT}\big(u_j\big)\rangle.\end{aligned}$$

If $\gamma < \frac{1}{1-p_1}$, by following a similar procedure as in Lemma 4.2.4 we can instead bound the terms in question from below, i.e.

$$\begin{aligned}\langle u_i, \mathrm{ATT}(u_j)\rangle &= \langle u_i, \frac{1}{Z_i}\sum_{j=1}^{n} e^{s_{jk}} u_k\rangle \\ &\geq \frac{1}{Z_i}\left(p_1 e^{\beta} + p_1 \sum_{j\neq i} e^{s_{ij}}\right) \\ &= p_1 + o_n(1),\end{aligned}$$

and

$$\begin{aligned}\langle \mathrm{ATT}(u_i), \mathrm{ATT}(u_j)\rangle &= \langle \frac{1}{Z_i}\sum_{k=1}^{n} e^{s_{ik}} u_k, \frac{1}{Z_i}\sum_{l=1}^{n} e^{s_{jl}} u_l\rangle \\ &\geq \frac{1}{Z_i Z_j}\left(p_1 e^{2\beta} + p_1 e^{\beta}\sum_{k\neq i} e^{s_{ik}} + p_1 e^{\beta}\sum_{l\neq j} e^{s_{jl}} + p_1 \sum_{k\neq i}\sum_{l\neq i} e^{s_{ik}+s_{jl}}\right) \\ &= p_1 + o_n(1).\end{aligned}$$

Hence,

$$\begin{aligned}\langle x_i', x_j'\rangle &\geq \alpha^2 \|x_i\|\|x_j\| p_1 + \alpha\big(\|x_i\| + \|x_j\|\big)p_1 + p_1 + o_n(1) \\ &= p_1(\alpha\|x_i\| + 1)\big(\alpha\|x_j\| + 1\big) + o_n(1).\end{aligned}$$

If $\gamma > \frac{1}{1-p_2}$, Lemma 4.2.3 can be applied to obtain

$$\begin{aligned}\langle x_i', x_j'\rangle &= \alpha^2 \|x_i\|\|x_j\|\langle u_i, u_j\rangle + \alpha\|x_i\|\langle u_i, u_j + \boldsymbol{o}_n(1)\rangle \\ &+ \ \alpha\big\|x_j\big\|\langle u_j, u_i + \boldsymbol{o}_n(1)\rangle + \langle u_i + \boldsymbol{o}_n(1), u_j + \boldsymbol{o}_n(1)\rangle \\ &= (\alpha\|x_i\| + 1)\big(\alpha\|x_j\| + 1\big)\langle u_i, u_j\rangle + o_n(1).\end{aligned}$$

□

We now prove Theorem 4.2.1.

Proof of Theorem 4.2.1. First, consider the case where $\gamma < \frac{1}{1-p_1}$. We can further simplify the result in Lemma 4.2.4 by completing the square, giving

$$\begin{aligned}\|x_i'\|^2 &\leq \alpha^2\|x_i\|^2 + 2\alpha\|x_i\|p_2 + p_2 + o_n(1) \\ &= \alpha^2\|x_i\|^2 + 2\alpha\|x_i\| + 1 - 2\alpha\|x_i\| - 1 + 2\alpha\|x_i\|p_2 + p_2 + o_n(1) \\ &= (\alpha\|x_i\| + 1)^2 - (1 - p_2)(2\alpha\|x_i\| + 1) + o_n(1) \\ &\leq (\alpha\|x_i\| + 1)^2 - (1 - p_2)(2\alpha\sqrt{q_1} + 1) + o_n(1),\end{aligned}$$

where the last inequality follows from the fact that $\|x_i\| \geq \sqrt{q_1}$. Thus, there is some constant $\delta > 0$ depending on α, p_2, q_1, q_2 such that

$$\frac{1}{\|x_i\|} \geq \frac{1+\delta}{\alpha\|x_i\|+1} + o_n(1).$$

An equivalent formulation of the result of Lemma 4.2.5 is the equation

$$\langle u_i', u_j' \rangle \geq \frac{(\alpha\|x_i\|+1)(\alpha\|x_j\|+1)}{\|x_i\|\|x_j\|} p_1 + o_n(1),$$

and hence

$$\langle u_i', u_j' \rangle \geq p_1(1+\delta)^2 + o_n(1) = p_1 + \varepsilon + o_n(1),$$

where $\varepsilon = p_1(2\delta + \delta^2)$. Since this was for any $i \neq j$, it follows that

$$\liminf_{n\to\infty} \min_{i\neq j} \langle u_i', u_j' \rangle \geq p_1 + \varepsilon.$$

For the case where $\gamma > \frac{1}{1-p_2}$, we formulate the result of Lemma 4.2.5 as

$$\langle u_i', u_j' \rangle = \frac{(\alpha\|x_i\|+1)(\alpha\|x_j\|+1)}{\|x_i\|\|x_j\|} \langle u_i, u_j \rangle + o_n(1)$$

and substitute the result of Lemma 4.2.4 to immediately obtain

$$\langle u_i', u_j' \rangle = \langle u_i, u_j \rangle + o_n(1).$$

Hence,

$$\lim_{n\to\infty} \langle u_i', u_j' \rangle = \langle u_i, u_j \rangle.$$

□

Corollary 4.2.6 (Phase Transition, Simplex Case): Assume there exist constants $q > 0$, $p \in (0,1)$ such that $\|x_i\|^2 = q$ and $\langle u_i, u_j \rangle = p$ for any $i \neq j$. Then,

$$\lim_{n\to\infty} \langle u_i', u_j' \rangle = \begin{cases} \frac{p(\alpha\sqrt{q}+1)^2}{\alpha^2 q + 2\alpha\sqrt{q}p + p} & \gamma < \frac{1}{1-p}, \\ \frac{p(\alpha\sqrt{q}+1)^2}{\alpha^2 q + \alpha\sqrt{q}(1+p) + \frac{1+3p}{4}} & \gamma = \frac{1}{1-p}, \\ p & \gamma > \frac{1}{1-p}. \end{cases}$$

Proof. This corresponds to the special case of Theorem 4.2.1 where $p_1 = p_2$.

If $\gamma < \frac{1}{1-p}$, the steps of the proof of Lemma 4.2.4 follow with equality, giving

$$\|x_i'\|^2 = \alpha^2 q + 2\alpha\sqrt{q}p + p + o_n(1).$$

Similar reasoning can be used with Lemma 4.2.5 to obtain

$$\langle x_i', x_j' \rangle = p(\alpha\sqrt{q}+1)^2 + o_n(1).$$

Hence,

$$\lim_{n\to\infty} \langle u_i', u_j' \rangle = \frac{p(\alpha\sqrt{q}+1)^2}{\alpha^2 q + 2\alpha\sqrt{q}p + p}.$$

If $\gamma = \frac{1}{1-p}$, we first observe

$$\frac{\sum_{j\neq i} e^{s_{ij}}}{e^\beta} = \frac{(n-1)n^{\gamma p}}{n^\gamma} = n^{\gamma(p-1)+1} - n^{\gamma(p-1)} = 1 + o_n(1),$$

thus $Z_i = (2 + o_n(1))e^\beta$. Similar calculations as in Lemma 4.2.4 yield

$$\langle u_i, \mathrm{ATT}(u_i) \rangle = \frac{1+p}{2} + o_n(1),$$
$$\langle \mathrm{ATT}(u_i), \mathrm{ATT}(u_i) \rangle = \frac{1+3p}{4} + o_n(1).$$

Similar calculations as in Lemma 4.2.5 give

$$\langle u_i, \mathrm{ATT}(u_j) \rangle = p + o_n(1),$$
$$\langle \mathrm{ATT}(u_i), \mathrm{ATT}(u_j) \rangle = p + o_n(1).$$

By substituting these results into the inner product expansions, we obtain

$$\langle u_i', u_j' \rangle = \frac{\langle x_i', x_j' \rangle}{\|x_i'\|\|x_j'\|} = \frac{p(\alpha\sqrt{q}+1)^2}{\alpha^2 q + \alpha\sqrt{q}(1+p) + \frac{1+3p}{4}} + o_n(1).$$

If $\gamma > \frac{1}{1-p}$, Theorem 4.2.1 directly gives

$$\lim_{n\to\infty} \langle u_i, u_j \rangle = \langle u_i, u_j \rangle = p.$$

□

Since $p < 1$, the trailing term in the denominators of the $\gamma \geq \frac{1}{1-p}$ cases show that we have $\lim_{n\to\infty} \langle u_i', u_j' \rangle > p$. Thus, even when scaled appropriately, attention is still inherently a contractive operator.

Remark 4.2.1: In the absence of residual connections, i.e. $\alpha = 0$, we have

$$\lim_{n\to\infty} \langle u_i', u_j' \rangle = \begin{cases} 1 & \gamma < \frac{1}{1-p}, \\ \frac{4p}{1+3p} & \gamma = \frac{1}{1-p}, \\ p & \gamma > \frac{1}{1-p}. \end{cases}$$

The phase transition is especially clear in this case. In the subcritical regime, all token embeddings collapse onto the same direction, which implies that attention is behaving like an averaging operator, $\mathrm{ATT}(u_i) = \frac{1}{n}\sum_j u_j$. In the supercritical regime, the geometry of the embeddings is asymptotically unchanged and attention behaves like the identity map. In some sense, this provides a reason for why residual connections are crucical in avoiding this rank collapse phenomenon.

4.3 Backpropagation

Theorem 4.3.1 (Phase Transition in Attention Gradient):

Assume there exist constants $q_1, q_2 > 0$ and $p_1, p_2 \in (0,1)$ such that $q_1 \leq \|x_i\|^2 \leq q_2$ and $p_1 \leq \langle u_i, u_j \rangle \leq p_2$ for any $i \neq j$, with $p_1 = \langle u_i, u_j \rangle$ for some i, j. Let $\beta = \gamma \log n$ with $\gamma > 0$.

1. If $\gamma < \frac{1}{1-p_1}$,

$$\frac{1}{nd}\|\nabla_X X'\|^2 \leq \frac{4\gamma^2 (\log n)^2}{q_1 d} + o_n(1).$$

2. If $\gamma > \frac{1}{1-p_2}$,

$$\frac{1}{nd}\|\nabla_X X'\|^2 \geq \frac{1}{q_2}\left(1 - \frac{1}{d}\right) + o_n(1).$$

[Che+25]

Lemma 4.3.2: For any $i, k \in \{1, ..., n\}$ and $u, w \in \{1, ..., d\}$,

$$\frac{\partial (N(x_k))_w}{\partial (x_i)_u} = \delta_{ik} \frac{\delta_{wu}\|x_k\|^2 - (x_k)_w (x_k)_u}{\|x_k\|^3}$$

Proof. By the quotient rule,

$$\frac{\partial (N(x_k))_w}{\partial (x_i)_u} = \frac{\partial \left((x_k)_w \cdot \|x_k\|^{-1} \right)}{\partial (x_i)_u} = \delta_{ik} \frac{\delta_{wu}\|x_k\| - (x_k)_w \cdot \frac{(x_k)_u}{\|x_k\|}}{\|x_k\|^2} = \delta_{ik} \frac{\delta_{wu}\|x_k\|^2 - (x_k)_w (x_k)_u}{\|x_k\|^3}.$$

□

Lemma 4.3.3: For any $k, j \in \{1, ..., n\}$ and $w, v \in \{1, ..., d\}$,

$$\begin{aligned}\frac{\partial \left(\mathrm{ATT}(u_j)\right)_v}{\partial (u_k)_w} = \Bigg[&\left(\delta_{kj} \beta \left(\sum_{m=1}^{n} e^{\beta\langle u_j, u_m \rangle} (u_m)_w (u_m)_v \right) + e^{\beta \langle u_j, u_k \rangle} \left(\beta (u_j)_w (u_k)_v + \delta_{wv} \right) \right) \cdot Z_j \\ - &\left(\delta_{kj} \beta \left(\sum_{l=1}^{n} e^{\beta \langle u_j, u_l \rangle} (u_l)_w \right) + \beta e^{\beta \langle u_j, u_k \rangle} (u_j)_w \right) \cdot \left(\sum_{m=1}^{n} e^{\beta \langle u_j, u_m \rangle} (u_m)_v \right) \Bigg] \cdot Z_j^{-2}\end{aligned}$$

Proof. Recall

$$\left(\mathrm{ATT}(u_j)\right)_v = \frac{\sum_{m=1}^{n} e^{\beta \langle u_j, u_m \rangle} (u_m)_v}{\sum_{l=1}^{n} e^{\beta \langle u_j, u_l \rangle}} := \frac{(y_j)_v}{Z_j}$$

The derivative of the numerator,

$$\begin{aligned}\frac{\partial\left(y_j\right)_v}{\partial\left(u_k\right)_w} &= \sum_{m=1}^{n}\left[\left(\frac{\partial}{\partial\left(u_k\right)_w} e^{\beta\langle u_j, u_m\rangle}\right)\left(u_m\right)_v + e^{\beta\langle u_j, u_m\rangle}\frac{\partial}{\partial\left(u_k\right)_w}\left(u_m\right)_v\right] \\ &= \sum_{m=1}^{n}\left[\beta\left(\delta_{kj}\left(u_m\right)_w + \delta_{km}\left(u_j\right)_w\right)e^{\beta\langle u_j, u_m\rangle}\left(u_m\right)_v + e^{\beta\langle u_j, u_m\rangle}\left(\delta_{km}\delta_{wv}\right)\right] \\ &= \left(\beta\delta_{kj}\sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}\left(u_m\right)_w\left(u_m\right)_v\right) + e^{\beta\langle u_j, u_k\rangle}\left(\beta\left(u_j\right)_w\left(u_k\right)_v + \delta_{wv}\right).\end{aligned}$$

The derivative of the denominator,

$$\frac{\partial Z_j}{\partial\left(u_k\right)_w} = \sum_{l=1}^{n}\beta\left(\delta_{kj}\left(u_l\right)_w + \delta_{kl}\left(u_j\right)_w\right)e^{\beta\langle u_j, u_l\rangle} = \left(\delta_{kj}\beta\sum_{l=1}^{n} e^{\beta\langle u_j, u_l\rangle}\left(u_l\right)_w\right) + \beta e^{\beta\langle u_j, u_k\rangle}\left(u_j\right)_w.$$

Applying the quotient rule yields the desired result. □

Lemma 4.3.4: The Jacobian of the attention operator has the form

$$\left(\frac{\partial\left(\left(\mathrm{ATT}\left(N\left(x_j\right)\right)\right)_v\right)}{\partial\left(x_i\right)_u}\right)_{u,v=1}^{d} = \left\|x_i\right\|^{-1/2}\left[\left(\mathbf{R}_1 + \mathbf{R}_2\right)Z_j - \left(\mathbf{U_1} + \mathbf{U_2}\right)\otimes\mathbf{V}_j\right]\cdot Z_j^{-2},$$

where

$$\begin{aligned}&\mathbf{R}_1 = \delta_{ij}\left(\mathbf{W}_j - u_i\otimes\left(\mathbf{W}_j u_j\right)\right), && \mathbf{R}_2 = e^{\beta\langle u_j, u_i\rangle}\left(\left(-u_j + \beta\mathbf{P}_{u_i}u_j\right)\otimes u_i + I_d\right), \\ &\mathbf{U}_1 = \delta_{ij}\beta\left(\mathbf{P}_{u_i}\mathbf{V}_j\right), && \mathbf{U}_2 = \beta e^{\beta\langle u_j, u_i\rangle}\left(\mathbf{P}_{u_i}u_j\right), \\ &\mathbf{V}_j = \sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}u_m, && \mathbf{W}_j = \sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}u_m\otimes u_m, \\ &\mathbf{P}_x y = y - \langle y, x\rangle x.\end{aligned}$$

Proof. Let

$$J_{wv} = \frac{\partial\left(\mathrm{ATT}\left(N\left(x_j\right)\right)\right)_v}{\partial\left(N\left(x_k\right)\right)_w} = \frac{\partial\left(\mathrm{ATT}\left(u_j\right)\right)_v}{\partial\left(u_k\right)_w},$$

which was computed in the previous lemma. By the multivariate chain rule, we have

$$\begin{aligned}
\frac{\partial\Big(\big(\mathrm{ATT}(N(x_j))\big)_v\Big)}{\partial(x_i)_u} &= \sum_{k=1}^{n}\sum_{w=1}^{d} J_{wv}\cdot\frac{\partial\Big(N(x_k)_w\Big)}{\partial(x_i)_u} \\
&= \sum_{w=1}^{d} J_{wv}\cdot\frac{\delta_{wu}\|x_i\|^2-(x_i)_w(x_i)_u}{\|x_i\|^3} \\
&= \|x_i\|^{-1}\left(J_{uv}-(u_i)_u\sum_{w=1}^{d} J_{wv}(u_i)_w\right).
\end{aligned}$$

We compute

$$\begin{aligned}
&\sum_{w=1}^{d} J_{wv}(u_i)_w \\
&= [\left(\delta_{ij}\beta\left(\sum_{m=1}^{n} e^{\beta\langle u_j,u_m\rangle}\langle u_m,u_i\rangle(u_m)_v\right)+e^{\beta\langle u_j,u_i\rangle}\Big(\big(\beta\langle u_j,u_i\rangle+1\big)(u_i)_v\Big)\cdot Z_j\right. \\
&- \quad\left.\left(\delta_{ij}\beta\left(\sum_{l=1}^{n} e^{\beta\langle u_j,u_l\rangle}\langle u_l,u_i\rangle\right)+\beta e^{\beta\langle u_j,u_i\rangle}\langle u_j,u_i\rangle\right)\cdot\left(\sum_{m=1}^{n} e^{\beta\langle u_j,u_m\rangle}(u_m)_v\right)\right]\cdot Z_j^{-2},
\end{aligned}$$

which then gives

$$\begin{aligned}
\frac{\partial\Big(\big(\mathrm{ATT}(N(x_j))\big)_v\Big)}{\partial(x_i)_u} = &\left[\left(\delta_{ij}\beta\left(\sum_{m=1}^{n} e^{\beta\langle u_j,u_m\rangle}\Big((u_m)_u(u_m)_v-\langle u_m,u_i\rangle(u_m)_v(u_i)_u\Big)\right)\right.\right. \\
&+\quad \left. e^{\beta\langle u_j,u_k\rangle}\Big(\beta(u_j)_u(u_k)_v+\delta_{uv}-\big(\beta\langle u_j,u_i\rangle+1\big)(u_i)_v(u_i)_u\Big)\right)\cdot Z_j \\
&-\quad\left(\delta_{ij}\beta\left(\sum_{l=1}^{n} e^{\beta\langle u_j,u_l\rangle}\Big((u_l)_u-\langle u_l,u_i\rangle(u_i)_u\Big)\right)\right. \\
&+\quad\left.\left.\beta e^{\beta\langle u_j,u_k\rangle}\Big((u_j)_u-\langle u_j,u_i\rangle(u_i)_u\Big)\right)\cdot\left(\sum_{m=1}^{n} e^{\beta\langle u_j,u_m\rangle}(u_m)_v\right)\right] \\
&\cdot Z_j^{-2}\cdot\|x_i\|^{-1}.
\end{aligned}$$

When written in matrix form (i.e. the transposed Jacobian), this becomes

$$\Bigg[\Big[\delta_{ij}\beta\big(\mathbf{W}_j - u_i \otimes (W_j u_i)\big) + e^{\beta\langle u_j, u_k\rangle}\big(\beta u_j \otimes u_i + I_d - \big(\beta\langle u_j, u_i\rangle + 1\big)\, u_i \otimes u_i\big)\Big] \cdot Z_j$$

$$-\Big[\delta_{ij}\beta\big(\mathbf{V}_j - \langle \mathbf{V}_j, u_i\rangle u_i\big) + \beta e^{\beta\langle u_j, u_i\rangle}\big(u_j - \langle u_j, u_i\rangle u_i\big)\big)\Big] \otimes \mathbf{V}_j\Bigg] \cdot Z_j^{-2} \cdot \|x_i\|^{-1}$$

$$= \Bigg[\Big(\mathbf{R}_1 + e^{\beta\langle u_j, u_i\rangle}\Big(\big(-u_i + \beta \mathbf{P}_{u_i} u_j\big) \otimes u_i + I_d\Big)\Big) \cdot Z_j$$

$$-\Big[\delta_{ij}\beta\Big(\mathbf{P}_{u_i}\mathbf{V}_j\Big) + \beta e^{\beta\langle u_j, u_i\rangle}\Big(\mathbf{P}_{u_i} u_j\Big)\Big] \otimes \mathbf{V}_j\Bigg] \cdot Z_j^{-2} \cdot \|x_i\|^{-1}$$

$$= \Bigg[(\mathbf{R}_1 + \mathbf{R}_2) \cdot Z_j + (\mathbf{U}_1 + \mathbf{U}_2) \otimes \mathbf{V}_j\Bigg] \cdot Z_j^{-2} \cdot \|x_i\|^{-1},$$

where the intermediate steps follow from basic properties of the tensor product. □

We now prove Theorem 4.2.1.

Proof of Theorem 4.2.1. Throughout this proof, we use the fact that $\|x \otimes y\|_F = \|x\|_2\|y\|_2$, as well as $\|A\|_F^2 = \mathrm{tr}(A^T A)$. The subscripts will be dropped since we are exclusively using the Frobenius norm for matrices and 2-norm for vectors. First consider the case when $\gamma > \frac{1}{1-p_2}$.

1. When $i \neq j$, $\mathbf{R}_1 Z_j^{-1} = 0$. When $i = j$, we obtain (using the form of $\mathbf{R}_1$ before simplification in the previous lemma)

$$\begin{aligned}
\|\mathbf{R}_1\| Z_j^{-1} &= \left\| \beta \sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}(u_m \otimes u_m - \langle u_m, u_i\rangle u_i \otimes u_m)\right\| \cdot Z_j^{-1} \\
&= \left\| \beta \sum_{m\neq j} e^{\beta\langle u_j, u_m\rangle}(u_m \otimes u_m - \langle u_m, u_i\rangle u_i \otimes u_m)\right\| \cdot Z_j^{-1} \\
&\leq \beta\left[\sum_{m\neq j} e^{\beta p_2}\Big(\|u_m\|^2 + \|u_m\|^2\|u_i\|^2\Big)\right] \cdot Z_j^{-1} \\
&= 2\beta(n-1)e^{\beta p_2} \cdot Z_j^{-1} \\
&\leq 2\gamma\log(n) \cdot n^{\gamma p_2 + 1 - \gamma}(1 + o_n(1))
\end{aligned}$$

where the last inequality follows from Lemma 4.2.2. Since $\gamma p_2 + 1 - \gamma < 0$ by assumption, this value goes to 0 as $n \to \infty$.

2. When $i = j$, we notice that $\mathbf{P}_{u_i} u_j = 0$, hence

$$\mathbf{R}_2 Z_j^{-1} = e^{\beta} Z_j^{-1}(-u_i \otimes u_i + I_d) = (I_d - u_i \otimes u_i)(1 + o_n(1))$$

When $i \neq j$, naively bounding gives

$$\begin{aligned}\|\mathbf{R}_2\|Z_j^{-1} &\leq e^{\beta p_2}Z_j^{-1}\left(\|u_i\|^2+\beta\|\mathbf{P}_{u_i}u_j\|\|u_i\|+\|I_d\|\right)\\ &\leq n^{\gamma(p_2-1)}\left(1+2\gamma\log n+\sqrt{d}\right)(1+o_n(1)),\end{aligned}$$

which goes to 0 as $n\to\infty$ since $\gamma>0$ and $p_2-1<0$.

3. When $i=j$, we have that $\mathbf{U}_2=0$ since $\mathbf{P}_{u_i}u_i=0$, so

$$\begin{aligned}&\left\|(\mathbf{U}_1+\mathbf{U}_2)\otimes\mathbf{V}_j\right\|\cdot Z_j^{-2}=Z_j^{-2}\beta\left\|\mathbf{P}_{u_i}\mathbf{V}_j\right\|\left\|\mathbf{V}_j\right\|\\ &\leq Z_j^{-2}\beta\left(\sum_{m\neq j}e^{\beta\langle u_j,u_m\rangle}\left\|\mathbf{P}_{u_i}u_m\right\|\right)\cdot\left(e^\beta+\sum_{m\neq j}e^{\beta\langle u_j,u_m\rangle}\|u_m\|\right)\\ &\leq 2e^{-2\beta}\beta\left(ne^{\beta p_2}\right)\left(e^\beta+ne^{\beta p_2}\right)(1+o_n(1))\\ &\leq 2\gamma\log n\cdot n^{\gamma(p_2-1)+1}\cdot\left(1+n^{\gamma(p_2-1)+1}\right)(1+o_n(1)),\end{aligned}$$

which goes to 0 as $n\to\infty$ since $\gamma(p_2-1)+1<0$. When $i\neq j$, $\mathbf{U_1}=0$, thus

$$\begin{aligned}&\left\|(\mathbf{U}_1+\mathbf{U}_2)\otimes\mathbf{V}_j\right\|\cdot Z_j^{-2}=Z_j^{-2}\beta e^{\beta\langle u_j,u_i\rangle}\left\|\mathbf{P}_{u_i}u_j\right\|\left\|\mathbf{V}_j\right\|\\ &\leq 2Z_j^{-2}\beta e^{\beta p_2}\sum_{m=1}^{n}e^{\beta\langle u_j,u_m\rangle}\|u_m\|\\ &\leq 2\beta e^{-2\beta}e^{\beta p_2}ne^{\beta}(1+o_n(1))\\ &\leq 2\gamma\log(n)n^{\gamma(p_2-1)+1}(1+o_n(1)),\end{aligned}$$

which goes to 0 as $n\to\infty$ since $\gamma(p_2-1)+1<0$. Combining all cases for $\gamma>\frac{1}{1-p_2}$, we have that

$$\begin{aligned}\left(\frac{\partial\left(\left(\mathrm{ATT}\left(N\left(x_j\right)\right)\right)_v\right)}{\partial\left(x_i\right)_u}\right)_{u,v=1}^{d}&\leq\frac{\delta_{ij}}{\|x_i\|}(I_d-u_i\otimes u_i)(1+o_n(1))\\ &=\frac{\delta_{ij}}{\|x_i\|}(I_d-u_i\otimes u_i)+\mathbf{o}_n(1)+o_n(1)I_d.\end{aligned}$$

Noticing the block-diagonal structure of $\nabla_X X'$ and that $(u_i\otimes u_i)^2=u_i\otimes u_i$,

$$\begin{aligned}
\frac{1}{nd}\|\nabla_X X'\|^2 &= \frac{1}{nd}\left(\sum_{i=1}^{n}\frac{1}{\|x_i\|}\|I_d - u_i \otimes u_i\|^2\right) + o_n(1) \\
&\geq \frac{1}{ndq_2}\left(\sum_{i=1}^{n}\operatorname{tr}(I_d - u_i \otimes u_i)\right) + o_n(1) \\
&= \frac{1}{ndq_2}\left(\sum_{i=1}^{n} d - \operatorname{tr}(u_i u_i^T)\right) + o_n(1) \\
&= \frac{1}{q_2}\left(1 - \frac{1}{d}\right) + o_n(1).
\end{aligned}$$

Next, consider the case when $\gamma < \frac{1}{1-p_2}$.

1. Once again, if $i \neq j$, $\mathbf{R_1} Z_j^{-1} = 0$. When $i = j$, we first note that $\left\|\mathbf{P}_{u_i} u_m\right\| \leq 1$, since $\mathbf{P}_{u_i} u_m$ gives the component of u_m orthogonal to u_i. Thus,

$$\begin{aligned}
\mathbf{R}_1 Z_j^{-1} &= \left\|\beta \sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}(u_m \otimes u_m - \langle u_m, u_i\rangle u_i \otimes u_m)\right\| \cdot Z_j^{-1} \\
&= \beta\left\|\sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}\left(\mathbf{P}_{u_i} u_m\right) \otimes u_m\right\| \cdot Z_j^{-1} \\
&\leq \beta Z_j^{-1}\sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}\left\|\mathbf{P}_{u_i} u_m\right\|\|u_m\| \\
&\leq \beta
\end{aligned}$$

2. As computed earlier, $\|I_d - u_i \otimes u_i\| = \sqrt{d-1}$, giving

$$\begin{aligned}
\|\mathbf{R}_2\| Z_j^{-1} &\leq Z_j^{-1} e^{\beta\langle u_j, u_i\rangle}\left\|\left(-u_i + \beta \mathbf{P}_{u_i} u_j\right) \otimes u_i + I_d\right\| \\
&\leq Z_j^{-1} e^{\beta\langle u_j, u_i\rangle}\left(\|I_d - u_i \otimes u_i\| + \beta\left\|\mathbf{P}_{u_i} u_j\right\|\|u_i\|\right) \\
&\leq Z_j^{-1} e^{\beta\langle u_j, u_i\rangle}\left(\sqrt{d-1} + \beta\right).
\end{aligned}$$

3. Note that $\left\|\mathbf{V}_j\right\| \leq Z_j$, and $\left\|\mathbf{P}_{u_i}\mathbf{V}_j\right\| \leq \left\|\mathbf{V}_j\right\| = Z_j$.

$$\begin{aligned}
\left\|(\mathbf{U}_1 + \mathbf{U}_2) \otimes \mathbf{V}_j\right\| \cdot Z_j^{-2} &\leq (\|\mathbf{U}_1\| + \|\mathbf{U}_2\|) \cdot Z_j^{-1} \\
&\leq Z_j^{-1}\delta_{ij}\beta\left\|\mathbf{P}_{u_i}\mathbf{V}_j\right\| + \beta e^{\beta\langle u_j, u_i\rangle}\left\|\mathbf{P}_{u_i} u_j\right\| \cdot Z_j^{-1} \\
&\leq \beta\left(\delta_{ij} + e^{\beta\langle u_j, u_i\rangle} Z_j^{-1}\right).
\end{aligned}$$

All together, this gives

$$
\begin{aligned}
&\left\| \left(\frac{\partial\Big(\big(\mathrm{ATT}(N(x_j))\big)_v \Big)}{\partial (x_i)_u} \right)_{u,v=1}^{d} \right\| \\
&\leq \frac{1}{\|x_i\|}\Big(\delta_{ij}\beta + Z_j^{-1} e^{\beta\langle u_j, u_i\rangle}\Big(\sqrt{d-1}+\beta\Big) + \beta\Big(\delta_{ij} + e^{\beta\langle u_j, u_i\rangle} Z_j^{-1}\Big)\Big) \\
&= \frac{1}{\|x_i\|}\Big(2\beta\delta_{ij} + \Big(2\beta + \sqrt{d}\Big) e^{\beta\langle u_j, u_i\rangle} Z_j^{-1}\Big).
\end{aligned}
$$

Using the properties of Z_j as shown in Lemma 4.2.2,

$$
\begin{aligned}
&\frac{1}{nd}\|\nabla_X X'\|^2 \\
&\leq \frac{1}{nd}\sum_{i,j=1}^{n} \frac{1}{\|x_i\|}\left(4\beta^2\delta_{ij} + 4\beta\delta_{ij}\Big(2\beta+\sqrt{d}\Big)\frac{e^{\beta\langle u_j,u_i\rangle}}{Z_j} + \Big(2\beta+\sqrt{d}\Big)^2 \cdot \frac{e^{2\beta\langle u_j,u_i\rangle}}{Z_j^2}\right) \\
&\leq \frac{1}{ndq_1}\left(n\cdot 4\beta^2 + O\big((\log n)^2\big)\left(\sum_{j=1}^{n}\frac{e^{\beta}}{Z_j}\right) + O\big((\log n)^2\big)\left(\sum_{i,j=1}^{n}\frac{e^{2\beta\langle u_j,u_i\rangle}}{Z_j^2}\right)\right) \\
&\leq \frac{(1+o_n(1))}{ndq_1}\left(4n\beta^2 + O\big((\log n)^2\big)\frac{n^{\gamma+1}}{n^{\gamma p_1+1}} + O\big((\log n)^2\big)\cdot\sum_{j=1}^{n}\left(\frac{n^{\gamma}}{n^{\gamma p_1+1}}\cdot\sum_{i=1}^{n}\frac{e^{\beta\langle u_j,u_i\rangle}}{Z_j}\right)\right) \\
&= \frac{(1+o_n(1))}{dq_1}\left(4\beta^2 + O\big((\log n)^2\big)n^{\gamma(1-p_1)-1} + O\big((\log n)^2\big)n^{\gamma(1-p_1)-1}\right) \\
&= 4\frac{\gamma^2(\log n)^2}{dq_1} + o_n(1),
\end{aligned}
$$

since $\gamma(1-p_1)-1<0$. □

5 Attention at Initialization

A natural next step after studying the equicorrelated setting is to consider the fully random case, where the inputs no longer share a common correlation structure and instead behave like independent high-dimensional noise. This serves as a baseline model in which any structure in the attention scores comes purely from random fluctuations, most importantly in the case of a newly initialized, pre-trained model. The purpose of this regime is not to improve upon the equicorrelated analysis, but to provide a contrasting basline.

5.1 Independent Entries

To identify a potential scaling for the full model, we first restrict ourselves to the simpler independent case. Consider a score vector $s = (s_1, ..., s_n)$ with independent and identically distributed entries $s_i \sim \mathcal{N}(0,1)$. If s_m is the maximum score, the maximum attention weight is given by

$$a_m = \frac{e^{\beta s_m}}{\sum_{k=1}^{n} e^{\beta s_k}} = \left(1 + \frac{\sum_{k \neq m} e^{\beta s_k}}{e^{s_m}}\right)^{-1}$$

For large n, we expect

$$\sum_{k \neq m} e^{\beta s_k} \sim n\mathbb{E}\left[e^{\beta s_1}\right] = ne^{\beta^2/2} = \exp(\log n + \beta^2/2).$$

It can also be shown that $s_m \sim \sqrt{2 \log n}$ [LLR83] and hence $e^{\beta s_m} \sim e^{\beta\sqrt{2 \log n}}$. A scaling for which $a_m \in (0, 1)$ must then have

$$\log n + \frac{\beta^2}{2} = \beta\sqrt{2 \log n},$$

which yields $\beta_n = \sqrt{2 \log n}$ after solving. Although a rough asymptotic argument, this identifies $\beta_n \asymp \sqrt{\log n}$ as the critical scaling factor under this i.i.d. assumption.

5.2 Correlated Entries

We now generalize to a model that resembles the attention mechanism more closely. Consider the $n \times n$ score matrix obtained from the multiplication

$$S = \left[\frac{(XW_Q)(XW_K)^T}{\sqrt{d}}\right],$$

where $X \in \mathbb{R}^{n \times d}$ is a random matrix with i.i.d. $\mathcal{N}(0, 1)$ components, and $W_Q, W_K \in \mathbb{R}^{d \times d}$ are random matrices with i.i.d. $\mathcal{N}(0, 1/d)$ entries.

In this context, S is a random object representing what score values may be produced by a randomly initialized model prior to training. For this reason, the score scaling β may equivalently be viewed as a scaling factor for the variance of i.i.d. $\mathcal{N}(0, \beta/d)$ entries in the W_Q and W_K matrices. This analysis is therefore also particularly useful in the effort of counteracting pathological behaviour at the beginning of training, which could in the worst case lead to an untrainable model. Since the methods used in this section are closely related to those of statistical physics, we will rely on asymptotic approximations and high-probability concentration results, as opposed to the exact results found in other sections.

Definition 5.2.1: Given attention weights $a_i \in \mathbb{R}^n$, define the *inverse participation ratio*

$$Y_i^{(2)} = \sum_{j=1}^{n} a_{ij}^2.$$

The inverse participation ratio heuristically satisfies

$$Y_i^{(2)} = \frac{1}{\text{effective support size of } a_i},$$

since $Y_i^{(2)} = 1$ if a_i has a single coordinate with value 1, and $Y_i^{(2)} = \frac{1}{n}$ if a_i is uniform. We are specifically interested in a scaling β that avoids these two cases in the large-n limit.

Theorem 5.2.2 (Phase Transition for IPR):

Under the scaling $\beta_n = \gamma\sqrt{\log n}$ where $\gamma > 0$ is a constant, the inverse participation ratio approximately satisfies

$$\lim_{n\to\infty} \mathbb{E}\left[Y_i^{(2)}(\beta)\right] = \begin{cases} 0, & \gamma < \sqrt{2}, \\ 1 - \frac{\sqrt{2}}{\gamma}, & \gamma > \sqrt{2} \end{cases}$$

given that the embedding dimension d is sufficiently large. Thus, under this model, the critical scaling is of order $\beta \asymp \sqrt{\log n}$ with phase transition at $\beta = \sqrt{2\log n}$. [GG25]

Lemma 5.2.3 (Concentration of Token Similarity): For two rows x_i, x_j of the embedding matrix X, define the *token similarity* $q_{ij} = \langle x_i, x_j\rangle / d$. Then,

$$q_{ij} \xrightarrow{p} \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

In particular, we may assume approximate deterministic equality for large d, such as those used in modern language models.

Proof. For $i = j$, we have

$$q_{ii} = \frac{1}{d}\sum_{k=1}^{d} (x_i)_k^2.$$

By the Law of Large Numbers, $q_{ii} \xrightarrow{p} \mathbb{E}\left[(x_i)_1^2\right] = 1$, since $(x_i)_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. For $i \neq j$,

$$\mathbb{E}\left[q_{ij}\right] = \frac{1}{d}\sum_{k=1}^{d} \mathbb{E}\left[(x_i)_k (x_j)_k\right] = 0,$$

and

$$\operatorname{Var}(q_{ij}) = \frac{1}{d} \cdot \mathbb{E}\left[(x_i)_k^2 (x_j)_k^2\right] = \frac{1}{d}.$$

By Chebyshev's Inequality, $q_{ij} \to 0$. Moreover, sharper concentration results (see [Ver26]) imply that these quantities are tightly concentrated with high probability for large d. □

Lemma 5.2.4 (Stein's Lemma): Let $Z = (Z_1, ..., Z_n)^T \sim \mathcal{N}(0, V)$ and $g : \mathbb{R}^n \to \mathbb{R}$ such that $g \in C^1$ and $\mathbb{E}[\|g(Z)\|], \mathbb{E}[\|\nabla g(Z)\|] < \infty$. Then,

$$\mathbb{E}[Z_i\, g(Z)] = \sum_{j=1}^{n} \operatorname{Cov}(Z_i, Z_j) \cdot \mathbb{E}\left[\frac{\partial}{\partial Z_j} g(Z)\right]$$

Proof. The pdf of Z has the form $p(z) = C \exp\left[-\frac{1}{2} z^T V^{-1} z\right]$. Thus,

$$\begin{aligned} \nabla p(z) &= -V^{-1} z p(z) \\ \Rightarrow z p(z) &= -V \nabla p(z) \\ \Rightarrow z_i p(z) &= -\sum_{j=1}^{n} V_{ij} \frac{\partial}{\partial z_j} p(z). \end{aligned}$$

Substituting the above equation into the calculation of expectation,

$$\mathbb{E}[Z_i g(Z)] = \int_{\mathbb{R}^n} z_i g(z) p(z) \, dz = -\sum_{j=1}^{n} V_{ij} \int_{\mathbb{R}^n} g(z) \frac{\partial}{\partial z_j} p(z) \, dz.$$

We then apply integration by parts along the dimension of z_j, noting that $g(z)$ vanishes when $z_j \to \pm\infty$ by assumption of finite expectation. This gives

$$\mathbb{E}[Z_i g(Z)] = \sum_{j=1}^{n} V_{ij} \int_{\mathbb{R}^n} \frac{\partial}{\partial z_j} g(z) p(z) \, dz = \sum_{j=1}^{n} \mathrm{Cov}\left(Z_i, Z_j\right) \cdot \mathbb{E}\left[\frac{\partial}{\partial z_j} g(Z)\right].$$

$\square$

Lemma 5.2.5 (Derivative of Scaled Softmax): Let $S \in \mathbb{R}^{n \times n}$. For any $i, j, k \in \{1, ..., n\}$,

$$\frac{\partial a_{ij}}{\partial s_{ik}} = \frac{\partial}{\partial s_{ik}} \left(\frac{e^{h\beta s_{ij}}}{\sum_{l=1}^{n} e^{h\beta s_{il}}} \right) = h\beta a_{ij} \left(\delta_{jk} - a_{ik}\right).$$

Proof. The derivative of the numerator is

$$\frac{\partial}{\partial s_{ik}} e^{h\beta s_{ij}} = h\beta e^{h\beta s_{ij}} \delta_{jk}.$$

The derivative of the denominator Z_i is

$$\frac{\partial}{\partial s_{ik}} Z_i = h\beta e^{h\beta s^{ik}}.$$

By quotient rule,

$$\frac{\partial}{\partial s_{ik}} \left(\frac{e^{h\beta s_{ij}}}{\sum_{l=1}^{n} e^{h\beta s_{il}}} \right) = \frac{h\beta e^{h\beta s_{ij}} \delta_{jk} Z - e^{h\beta s_{ij}} h\beta e^{h\beta s^{ik}}}{Z^2} = h\beta a_{ij} \left(\delta_{jk} - a_{ik}\right).$$

$\square$

Lemma 5.2.6: For large d, we have

$$s_{ij} \sim \mathcal{N}(0,1),$$

with covariance structure

$$\mathrm{Cov}\left(s_{ij}, s_{pq}\right) = q_{ip}q_{jq}.$$

Proof. By definition,

$$\begin{aligned}
s_{ij} &= \frac{1}{\sqrt{d}} \sum_{k=1}^{d} \left(W_Q x_i\right)_k \left(W_k x_j\right)_k \\
&= \frac{1}{\sqrt{d}} \sum_{k=1}^{d} \sum_{a,b=1}^{d} \left(W_Q\right)_{ka} \left(x_i\right)_a \left(W_K\right)_{kb} \left(x_j\right)_b \\
&:= \frac{1}{\sqrt{d}} \sum_{k=1}^{d} D_k.
\end{aligned}$$

First, $\mathbb{E}[D_k] = 0$ since all entries of W_K and W_Q have expectation 0. As for variance,

$$\begin{aligned}
\mathbb{E}\left[D_k^2\right] &= \mathbb{E}\left[\left(\sum_{a=1}^{d} \left(W_Q\right)_{ka} \left(x_i\right)_a\right)^2\right] \cdot \mathbb{E}\left[\left(\sum_{b=1}^{d} \left(W_K\right)_{kb} \left(x_j\right)_b\right)^2\right] \\
&= \left(\sum_{a=1}^{d} \mathbb{E}\left[\left(W_Q\right)_{ka}^2\right] \left(x_i\right)_a^2\right) \left(\sum_{b=1}^{d} \mathbb{E}\left[\left(W_K\right)_{ka}^2\right] \left(x_j\right)_b^2\right) \\
&= \frac{1}{d^2} \|x_i\|^2 \|x_j\|^2
\end{aligned}$$

We have that $\frac{\|x_k\|^2}{d} \xrightarrow{p} 1$ by Lemma 5.2.3, hence by the Central Limit Theorem, $s_{ij} \xrightarrow{d} \mathcal{N}(0,1)$. A similar calculation for covariance yields

$$\begin{aligned}
\mathrm{Cov}\left(s_{ij}, s_{pq}\right) &= \mathbb{E}\left[s_{ij} s_{pq}\right] \\
&= \frac{1}{d} \sum_{k,a,b,l,\alpha,\beta=1}^{d} \left(x_i\right)_a \left(x_j\right)_b \left(x_p\right)_\alpha \left(x_q\right)_\beta \mathbb{E}\left[\left(W_Q\right)_{ka} \left(W_Q\right)_{l\alpha}\right] \mathbb{E}\left[\left(W_K\right)_{kb} \left(W_K\right)_{l\beta}\right] \\
&= \frac{1}{d} \sum_{k,a,b=1}^{d} \left(x_i\right)_a \left(x_j\right)_b \left(x_p\right)_a \left(x_q\right)_b \mathbb{E}\left[\left(W_Q\right)_{ka}^2\right] \mathbb{E}\left[\left(W_K\right)_{ka}^2\right] \\
&= \frac{1}{d^3} \sum_{k=1}^{d} \langle x_i, x_p \rangle \langle x_j, x_q \rangle \\
&= q_{ip} q_{jq}.
\end{aligned}$$

$\square$

Lemma 5.2.7: Define the function

$$\Phi_n(\beta, h) = \mathbb{E}[\log Z_i(\beta, h)], \qquad Z_i(\beta, h) = \sum_{j=1}^{n} e^{h\beta s_{ij}}.$$

Then,

$$\lim_{n\to\infty} \mathbb{E}\left[Y_i^{(2)}\right] = 1 - \lim_{h\to 1} \lim_{n\to\infty} \frac{\partial}{\partial h} \frac{1}{\beta^2} \Phi_n(\beta, h).$$

Proof. We first note that we can safely swap derivative and expectation in the case of

$$\frac{\partial}{\partial h}\Phi_n(\beta, h) = \mathbb{E}\left[\frac{\partial}{\partial h} \log Z_i(\beta, h)\right] = \mathbb{E}\left[\frac{\sum_{j=1}^{n} \beta s_{ij} e^{h\beta s_{ij}}}{\sum_{k=1}^{n} e^{h\beta s_{ik}}}\right] = \beta \sum_{j=1}^{n} \mathbb{E}\left[s_{ij} a_{ij}\right],$$

where we temporarily use the notation $a_{ij} = \exp\left(h\beta s_{ij}\right)/Z_i(\beta, h)$. Since this is a sum of elements of the form $\mathbb{E}\left[s_{ij} g(s_i)\right]$ where $s_i \in \mathbb{R}^n$ is normally distributed, we apply Stein's Lemma to obtain

$$\begin{aligned}
\frac{\partial}{\partial h}\Phi_n(\beta, h) &= \beta \sum_{j=1}^{n} \sum_{k=1}^{n} \mathrm{Cov}\left(s_{ij}, s_{ik}\right) \mathbb{E}\left[\frac{\partial a_{ij}}{\partial s_{ik}}\right] \\
&= \beta^2 \sum_{j=1}^{n} \sum_{k=1}^{n} q_{ii} q_{jk} \mathbb{E}\left[h\beta a_{ij}\left(\delta_{jk} - a_{ik}\right)\right].
\end{aligned}$$

Define $Y_{i,h}^{(2)} = \sum_{j=1}^{n} a_{ij}^2$. By Lemma 5.2.3, $q_{ii} \xrightarrow{p} 1$ and $q_{ij} \xrightarrow{p} 0$ for $i \neq j$. This gives

$$\begin{aligned}
\frac{\partial}{\partial h}\Phi_n(\beta, h) &\approx \beta^2 h \sum_{j=1}^{n} \sum_{k=1}^{n} \delta_{jk} \mathbb{E}\left[a_{ij}\delta_{jk} - a_{ij} a_{ik}\right] \\
&= \beta^2 h \sum_{j=1}^{n} \mathbb{E}\left[a_{ij} - a_{ij}^2\right] \\
&= \beta^2 h \left(1 - \mathbb{E}\left[Y_{i,h}^{(2)}\right]\right).
\end{aligned}$$

Rearranging and taking limits gives

$$\lim_{n\to\infty} \mathbb{E}\left[Y_i^{(2)}\right] = 1 - \lim_{h\to 1} \lim_{n\to\infty} \frac{\partial}{\partial h} \frac{1}{\beta^2} \Phi_n(\beta, h).$$

□

We are now left with computing $\Phi_n(\beta, h) = \mathbb{E}[\log Z_i(\beta, h)]$. This is a problem similar to the well-studied Random Energy Model from statistical physics, for which methods exist. In particular, we use the replica method, which is based on the applicaiton of the limit

$$\mathbb{E}[\log Z] = \lim_{t\to 0^+} \frac{\mathbb{E}[Z^t] - 1}{t}.$$

In general, the replica method is not fully mathematically rigorous, especially in its use of analytic continuation and assumptions about the structure of overlap matrices. Despite this, it is able to provide accurate predictions in a wide range of disordered systems and high-dimensional probabilistic models.

Proof of Theorem 5.2.2. First, we consider the replicated system

$$\mathbb{E}[Z_i^t(\beta,h)] = \mathbb{E}\left[\left(\sum_{k=1}^{n} e^{h\beta s_{ik}}\right)\right]^t = \sum_{k_1,\ldots,k_t=1}^{n} \mathbb{E}\left[\exp\left(h\beta\sum_{a=1}^{t} s_{ik_a}\right)\right].$$

Using the moment-generating function of the multivariate normal distribution,

$$\begin{aligned}\mathbb{E}[Z_i^t(\beta,h)] &= \sum_{k_1,\ldots,k_t=1}^{n} \exp\left(\frac{h^2\beta^2}{2}\sum_{a,b=1}^{t}\mathbb{E}\left[s_{ik_a}s_{ik_b}\right]\right)\\ &= \sum_{k_1,\ldots,k_t=1}^{n} \exp\left(\frac{h^2\beta^2}{2}\sum_{a,b=1}^{t}\delta_{k_ak_b}\right)\end{aligned}$$

Define the empirical overlap matrix Q with entries $Q_{ab} = \delta_{k_ak_b}$. Among all n^t possible configurations of replica states, there are

$$S(Q) = \sum_{k_1,\ldots,k_t=1}^{n}\prod_{a,b=1}^{t}\mathbb{1}\left(Q_{ab} = \delta_{k_ak_b}\right)$$

with a specific overlap matrix Q. Thus,

$$\mathbb{E}[Z_i^t(\beta,h)] = \sum_{Q} S(Q)\exp\left(\frac{h^2\beta^2}{2}\sum_{a,b=1}^{t}Q_{ab}\right) = \sum_{Q}\exp\left(\log S(Q) + \frac{h^2\beta^2}{2}\sum_{a,b=1}^{t}Q_{ab}\right).$$

For large n, this sum is dominated by overlap structure Q that maximizes the exponent, in acordance with the Laplace principle. To evaluate this dominant contribution, we restrict attention to a class of structured overlap matrices, which is the *1-RSB Ansatz* step in the Replica method.

By a symmetric argument, we assume that for this maximizing Q, the t states across the replicas are divided into $\frac{t}{x}$ groups of x elements each, where replicas within the same group occupy the same state, i.e. $k_a = k_b$ if and only if replicas a and b are in the same group.

Although x will ulitimately be extended to real values via analytic continuation, we temporarily assume that relevent quantities are integers in order to analyze the system combinatorically. For any maximizing Q under this assumption,

$$S(Q) = n\cdot(n-1)\cdots\left(n-\frac{t}{x}+1\right)\cdot\frac{t!}{(x!)^{t/x}(t/x!)},$$

so $\log S(Q) \approx \frac{t}{x}\log n$ for large n, since $t, x \ll n$. Also,

$$\sum_{a,b=1}^{t} Q_{ab} = \sum_{\text{groups}} x^2 = tx,$$

which gives after Taylor expansion

$$\frac{\mathbb{E}[Z_i^t(\beta, h)] - 1}{t} = \frac{1}{x}\log n + \frac{h^2\beta^2}{2}x + O(t^2).$$

Determining the value of x corresponding to this maximizing Q reduces, in this representation, to a minimization over x, the reason for which lies beyond the scope of this derivation. In the analytic continuation with $t \to 0^+$, we minimize over $x \in (0, 1)$, giving

$$x = \min\left\{\frac{1}{h\beta}\sqrt{2\log n}, 1\right\}.$$

Using this,

$$\Phi_n(\beta, h) = \lim_{t\to 0^+}\left[\frac{\mathbb{E}[Z_i^t(\beta, h)] - 1}{t}\right] = \begin{cases} \log n + \frac{h^2\beta^2}{2}, & \beta < \frac{\sqrt{2\log n}}{h}, \\ h\beta\sqrt{2\log n}, & \beta > \frac{\sqrt{2\log n}}{h}. \end{cases}$$

We now assume $\beta_n = \gamma\sqrt{\log n}$. Thus,

$$\frac{\partial}{\partial h}\frac{1}{\beta^2}\Phi_n(\beta, h) = \begin{cases} h, & \gamma < \sqrt{2} \\ \frac{h\sqrt{2}}{\gamma}, & \gamma > \sqrt{2}. \end{cases}$$

By the previous lemma,

$$\lim_{n\to\infty} \mathbb{E}\left[Y_i^{(2)}(\beta)\right] = \begin{cases} 0, & \gamma < \sqrt{2}, \\ 1 - \frac{\sqrt{2}}{\gamma}, & \gamma > \sqrt{2}. \end{cases}$$

□

Corollary 5.2.8 (Update Average Cosine Similarity): Define the *updated token matrix*

$$Y = AXW_V,$$

where $A = \mathrm{softmax}(S)$, applied row-wise, and $W_V \in \mathbb{R}^{d\times d}$ is a random matrix with i.i.d. $\mathcal{N}(0, 1/d)$ entries. For sufficiently large embedding dimension d, the updated token similarity matrix approximately satisfies

$$\lim_{n\to\infty} \mathbb{E}\left[\frac{YY^T}{d}\right] = \begin{cases} 0, & \gamma < \sqrt{2}, \\ \left(1 - \frac{\sqrt{2}}{\gamma}\right)I_d, & \gamma > \sqrt{2}. \end{cases}$$

In other words, $\|y_i\|^2/d$

...

Proof. First observe that $\mathbb{E}[W_V W_V^T] = I_d$ by a simple calculation and for large d, $XX^T \approx I_n$ by Lemma 5.2.3. Since W_V and a_i are indepdendent,

$$\begin{aligned}
\mathbb{E}\left[\frac{\langle y_i, y_j \rangle}{d}\right] &= \frac{1}{d} \cdot \mathbb{E}\Big[(a_i X W_V)(a_j X W_V)^T\Big] \\
&= \frac{1}{d} \cdot \mathbb{E}\big[a_i X \, \mathbb{E}[W_V W_V^T] X^T a_j^T\big] \\
&\approx \mathbb{E}\big[a_i a_j^T\big] \\
= \delta_{ij} Y_i^2
\end{aligned}$$

□

6 Deterministic Score Theory

This section contains results found in [HK26].

6.1 Scaling Order

We now concern ourselves with a fixed score vector $s \in \mathbb{R}^n$. The following analysis is independent of the attention mechanism, in the sense that the scores are taken as given, where the specific token vectors and weight matrices used to obtain these scores are not of concern. Let s^* be the maximum of such scores, and define the *gaps* $\Delta_j = s^* - s_j$ from each score to the maximum. We refer to scores as *competitors*, as they are "competing" for the attention weight that is to be distributed by softmax. Thus, the attention weight of any given competitor can be rewritten as

$$a_j(\beta) = \frac{e^{-\beta \Delta_j}}{\sum_{k=1}^n e^{-\beta \Delta_k}},$$

which is equivalent to scaled softmax applied to the new score vector Δ with maximum 0. We define $Z(\beta) = \sum_{j=1}^n \exp(-\beta \Delta_j)$ as in previous sections, while noting that it is defined on gap scores and not raw scores in this context.

Definition 6.1.1 (CGF): The *cumulative gap-counting function* (CGF) is the number of competitors with gap at most t from the maximum,

$$N(t) = \sum_{j=1}^n \mathbb{1}(\Delta_j \le t).$$

Now, $Z(\beta)$ can be rewritten as the Laplace transform of the CGF, since

$$Z(\beta) = \sum_{j=1}^n e^{-\beta \Delta_j} = \sum_{j=1}^n \beta \int_0^\infty \mathbb{1}(\Delta_j \le t) e^{-\beta t} \, dt = \beta \int_0^\infty e^{-\beta t} N(t) \, dt.$$

This is clearly well-defined on finite n since $N(t) \le n$. However, we immediately observe that if $N(t)$ is not of exponential order, the attention weights tend toward uniformity.

Definition 6.1.2: The *upper tail accumulation scale* Λ is the smallest exponent for which $N(t) \leq \exp(\Lambda t)$ for all $t > 0$. Equivalently,

$$\Lambda = \sup_{t>0} \frac{\log N(t)}{t}.$$

When $\Lambda < \infty$, a gap Δ_j is called a *contact gap* if $N(\Delta_j) = e^{\Lambda \Delta_j}$. Let Δ_Λ be the largest of all such contact gaps and define the *contact accumulation exponent*

$$\alpha = \frac{\log N(\Delta_\Lambda)}{\log n} \in [0, 1],$$

which satisfies $N(\Delta_\Lambda) = n^\alpha$.

Figure 7 illustrates relevant quantities on sample data.

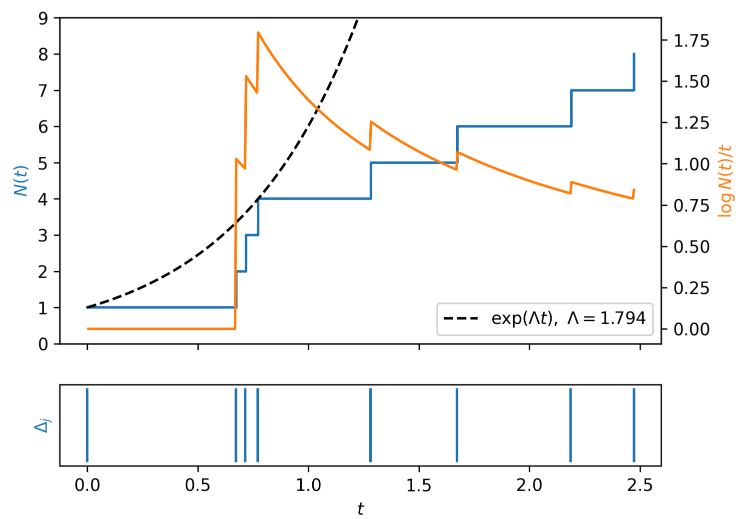

Figure 7: Visualization of gap counting quantities.

Lemma 6.1.3: The Shannon entropy $H(\beta) = -\sum_{j=1}^{n} a_j(\beta) \log(a_j(\beta))$ satisfies

$$H(\beta) = \log Z(\beta) - \beta \frac{Z'(\beta)}{Z(\beta)}$$

Proof. Recall that $\sum_{j=1}^{n} a_j(\beta) = 1$. Thus,

$$H(\beta) = -\sum_{j=1}^{n} a_j(\beta)\left(-\beta\Delta_j - \log Z(\beta)\right) = \log Z(\beta) + \sum_{j=1}^{n} \beta\Delta_j \frac{e^{-\beta\Delta_j}}{Z(\beta)} = \log Z(\beta) - \beta \frac{Z'(\beta)}{Z(\beta)}.$$

$\square$

Theorem 6.1.4 (Sequence Collapse Criterion):

For each $n \geq 2$, let $s^{(n)} \in \mathbb{R}^n$ be a score vector of length n with corresponding upper tail accumulation scale Λ_n. For any positive sequence (β_n),

1. If $\beta_n / \Lambda_n \to 0$, then *top-two collapse* holds, i.e.

$$\lim_{n\to\infty} D_n(\beta_n) = 0,$$

where $D_n(\beta_n)$ is the difference between the largest and second largest attention weights $a^{(n)}(\beta_n)$.

2. If $\beta_n / \Lambda_n \to \infty$, then *entropy collapse* holds, i.e.

$$\lim_{n\to\infty} H_n(\beta_n) = 0,$$

where is the Shannon entropy of the attention weights $a^{(n)}(\beta_n)$. [HK26]

Proof. Let $r_n = \beta_n / \Lambda_n$.

1. Let $\Delta_{\min}^{(n)}$ be the smallest non-zero gap, which necessarily has $\Delta_{\min}^{(n)} \leq \Delta_{\Lambda}^{(n)}$. The corresponding difference in attention weights can be bounded using $1 - e^{-x} \leq x$, giving

$$D_n(\beta_n) = \frac{1 - \exp\left(-\beta_n \Delta_{\min}^{(n)}\right)}{Z(\beta_n)} \leq \frac{\beta_n \Delta_{\min}^{(n)}}{Z(\beta_n)} \leq \frac{\beta_n \Delta_{\Lambda}^{(n)}}{Z(\beta_n)}.$$

By definition, we have that $\log\left(N\left(\Delta_{\Lambda}^{(n)}\right)\right) = \Lambda_n \Delta_{\Lambda}^{(n)} = \beta_n \Delta_{\Lambda}^{(n)} / r_n$. Substituting into and rearranging the definition of α_n gives $\beta_n \Delta_{\Lambda}^{(n)} = r_n \alpha_n \log n$. Since $e^{-\beta_n t}$ is monotone on the $N\left(\Delta_{\Lambda}^{(n)}\right)$ gaps with $\Delta_j \leq \Delta_{\Lambda}^{(n)}$, we have that

$$\begin{aligned} Z(\beta_n) &\geq N\left(\Delta_{\Lambda}^{(n)}\right) \exp\left(-\beta_n \Delta_{\Lambda}^{(n)}\right) \\ &= \exp\left(\beta_n \Delta_{\Lambda}^{(n)} / r_n\right) n^{r_n \alpha_n} \\ &= n^{\alpha_n (1 - r_n)}. \end{aligned}$$

Thus,

$$D_n(\beta_n) \leq \frac{\beta_n \Delta_{\Lambda}^{(n)}}{n^{\alpha_n (1 - r_n)}} = r_n \alpha_n \log n \cdot n^{-\alpha_n (1 - r_n)}.$$

Since $te^{-t} \leq e^{-1}$ for all $t \in \mathbb{R}$, setting $t = (1 - r_n) \cdot \alpha_n \log n$ yields

$$D_n(\beta_n) \leq \frac{r_n}{1 - r_n} \cdot e^{-1},$$

which goes to 0 as $n \to \infty$ since $r_n \to 0$ by assumption.

2. First observe that $Z(\beta_n) \geq 1$, since the gap between the maximum score and itself is always 0. Using the previously established integral form for $Z(\beta_n)$ as well as the fact that $N(t) \leq \exp\bigl(\Lambda^{(n)} t\bigr)$ by definition,

$$Z(\beta_n) = \beta_n \int_0^\infty e^{-\beta_n t} N(t)\, dt \leq \beta_n \int_0^\infty e^{-(\beta_n - \Lambda_n)t}\, dt = \frac{\beta_n}{\beta_n - \Lambda_n} = \frac{r_n}{r_n - 1},$$

which goes to 1 as $n \to \infty$ since $r_n \to \infty$. Thus, $Z(\beta_n) \to 1$.

For any n, differentiating the integral form with respect to β gives

$$Z'(\beta) = \int_0^\infty e^{-\beta t} N(t)\, dt + \beta \frac{d}{d\beta}\left(\int_0^\infty e^{-\beta t} N(t)\, dt\right) = \beta^{-1} Z(\beta) - \beta \int_0^\infty t e^{-\beta t} N(t)\, dt,$$

which implies that

$$\begin{aligned} -\beta Z'(\beta) &= \beta^2 \int_0^\infty t e^{-\beta t} N(t)\, dt - Z(\beta) \\ &\leq \beta^2 \int_0^\infty t e^{-(\beta - \Lambda_n)t}\, dt - Z(\beta) \\ &= \left(\frac{\beta}{\beta - \Lambda_n}\right)^2 - \left(\frac{\beta}{\beta - \Lambda_n}\right). \end{aligned}$$

Since $(\beta_n / (\beta_n - \Lambda_n)) \to 1$ and $-\beta Z'(\beta) = \beta \sum_{j=1}^n \Delta_j \exp\bigl(-\beta \Delta_j\bigr) \geq 0$, we have that $-\beta Z'(\beta) \to 0$. By Lemma 6.1.3,

$$\lim_{n\to\infty} H(\beta_n) = \lim_{n\to\infty} \left[\log Z(\beta_n) + \frac{1}{Z(\beta_n)}(-\beta(Z'(\beta)))\right] = 0.$$

$\square$

This establishes Λ_n as the order of the critical scaling for $s^{(n)}$. Denote $a_n \asymp b_n$ if there exists constants $c, C > 0$ such that for large enough n, $c b_n \leq a_n \leq C b_n$.

Corollary 6.1.5 (Critical Scaling Exponent): Let $\xi_\Lambda > 0$. If $\Lambda_n \asymp (\log n)^{\xi_\Lambda}$, then ξ_Λ is unique and any non-collapsing scaling β_n must have $\beta_n \asymp (\log n)^{\xi_\Lambda}$.

Furthermore, if $\alpha_n \asymp (\log n)^{\xi_\alpha}$ and $\Delta_\Lambda^{(n)} \asymp (\log n)^{\xi_\Delta}$, then

$$\xi_\Lambda = \xi_\alpha - \xi_\Delta + 1.$$

Proof. If $\Lambda_n \asymp (\log n)^{\xi'}$ for $\xi' > 0$, then $1 \asymp (\log n)^{\xi_\Lambda - \xi'}$, which is only possible when $\xi_\Lambda = \xi'$. For a non-collapsing scaling, we must have $\beta_n / \Lambda_n \asymp 1$, which is only possible if $\beta_n \asymp (\log n)^{\xi_\Lambda}$.

Since $\Lambda_n = (\alpha_n \log n) / \Delta_\Lambda^{(n)}$,

$$\Lambda_n \asymp \frac{(\log n)^{\xi_\alpha}(\log n)}{(\log n)^{\xi_\Delta}} = (\log n)^{\xi_\alpha - \xi_\Delta + 1},$$

and by uniqueness of this exponent must mean that $\xi_\Lambda = \xi_\alpha - \xi_\Delta + 1$. □

Consider the Simplex case from Section 4. For any row i of the $n \times n$ similarity matrix,

$$a_{ij} = \begin{cases} q, & i = j, \\ p, & i \neq j. \end{cases}$$

The gap counting function is given by

$$N(t) = \begin{cases} 1, & t < q - p, \\ n, & t \geq q - p \end{cases}$$

which immediately gives $\Lambda_n = (q-p)^{-1}\log(n)$, $\alpha_n = 1$, $\Delta_\Lambda^{(n)} = q - p$. Thus, $\xi_\Lambda = 1$, $\xi_\alpha = 0$, and $\xi_\Delta = 0$, which satisfy the relation derived in Corollary 6.1.5. Furthermore, we see that the non-collapsing scaling has $\beta_n \asymp \log n$, which validates Theorem 4.2.1.

6.2 Rank-Collapse Boundary

Proposition 6.2.1 (Ties at the maximum): Let $s \in \mathbb{R}^n$. If $N(0) \geq 2$, then $\Lambda = \infty$ and for every $\beta > 0$,

$$\|a(\beta)\|_\infty \leq \frac{1}{N(0)}, \quad H(\beta) \geq \log N(0).$$

In particular, for a sequence of score vectors $\left(s^{(n)}\right)$ with $s^{(n)} \in \mathbb{R}^n$ and $N(0) \geq 2$ for infinitely many n, no sequence X_n exists such that $\frac{\beta_n}{X_n} \to \infty$ implies that $H(\beta_n) \to 0$.

Proof. Since $N(0) \geq 2$ and N is non-decreasing, we have that

$$\Lambda = \sup_{t>0} \frac{\log N(t)}{t} \geq \sup_{t>0} \frac{\log 2}{t} = \infty.$$

The top $N(0)$ post-softmax weights are equal and have total mass at most 1, thus $\|a(\beta)\|_\infty \leq 1/N(0)$. This gives entropy bound

$$H(\beta) \geq \sum_{j=1}^{n} a_j(\beta)\log(N(0)) = \log N(0) \geq \log 2,$$

since post-softmax weights sum to 1. The conclusion for the sequence $\left(s^{(n)}\right)$ follows from the fact that $H(\beta) \geq \log 2$ for infinitely many n. □

Definition 6.2.2: Given a score vector $s \in \mathbb{R}^n$, define the *rank free energy* $\mathcal{F}(\beta) = \log Z(\beta)$ and the *maximum post-softmax gap* $\mathcal{G}(\beta) = \max_{j,k}|a_j(\beta) - a_k(\beta)|$.

Proposition 6.2.3 (Rank-Collapse Criterion): Let $\left(s^{(n)}\right)$ be a sequence of score vectors such that $s^{(n)} \in \mathbb{R}^n$. For any positive sequence (β_n), the following are equivalent:
1. $\mathcal{F}(\beta_n) \to \infty$,
2. $\mathcal{G}(\beta_n) \to 0$,
3. $\max_j |a_j^{(n)}(\beta_n) - 1/n| \to 0$.

Proof. For simplicity, we omit the argument β from the post-softmax weights $a_j^{(n)}(\beta)$. The maximum value of s has $\Delta_j = 0$, so

$$\max_j a_j^{(n)} = \frac{1}{Z(\beta)} = e^{-\mathcal{F}(\beta)}.$$

Since $\mathcal{G}(\beta) \leq \max_j a_j^{(n)}$, $\mathcal{F}(\beta_n) \to \infty$ implies that $\mathcal{G}(\beta_n) \to 0$.

Conversely, by a basic property of the arithmetic mean, we have the bound $\min_j a_j^{(n)}(\beta) \leq \frac{1}{n}$, which gives

$$\max_j a_j^{(n)} = \max_j a_j^{(n)} - \min_k a_k^{(n)} + \min_k a_k^{(n)} \leq \mathcal{G}(\beta) + \frac{1}{n}.$$

Thus, $\mathcal{G}(\beta_n) \to 0$ implies that $\max_j a_j^{(n)} \to 0$, which is equivalent to $\mathcal{F}(\beta) \to \infty$.

Finally, we observe that $\max_j a_j^{(n)} \to 0$ is equivalent to $\max_j |a_j^{(n)}(\beta_n) - 1/n| \to 0$, since $0 \leq a_k^{(n)} \leq \max_j a_j^{(n)}$ and $1/n \to 0$. □

6.3 Resolved Accumulation

Definition 6.3.1 (Rank Boundary): For $0 \leq r \leq \log n$, define

$$B(r) = \sup\{\beta \geq 0 : \mathcal{F}(\beta) \geq r\} = \sup\{\beta \geq 0 : Z(\beta) \geq e^r\}.$$

Definition 6.3.2 (Resolved Accumulation Scale): For $r \geq 0$ and a fixed $s \in \mathbb{R}^n$, define

$$\Lambda^{(r)} = \sup_{t>0:\, \log N(t)>r} \frac{\log N(t) - r}{t},$$

with convention $\sup \varnothing = 0$.

Proposition 6.3.3: Let (r_n) be a positive sequence such that $r_n \to \infty$. If $\beta_n < \Lambda_n^{(r_n)}$ eventually, then $\mathcal{G}(\beta_n) \to 0$.

Proof. By assumption, for large enough n we have $\beta_n < \Lambda_n^{(r_n)}$. Hence, there is some t_n such that

$$\beta_n < \frac{\log N(t_n) - r_n}{t_n},$$

since $\Lambda_n^{(r_n)}$ is defined in terms of a supremum. Rearranging,

$$N(t_n)e^{-\beta_n t_n} > e^{r_n}.$$

Since $N(t_n)$ competitors each contribute at least $e^{-\beta_n t_n}$ to $Z(\beta_n)$, we have that

$$Z(\beta_n) > e^{r_n},$$

which gives that $\mathcal{F}(\beta_n) \to \infty$, and thus $\mathcal{G}(\beta_n) \to 0$ by Proposition 6.2.3. □

Proposition 6.3.4: Define the Laplace envelope

$$S(\beta) = \sup_{t>0}\Big(\log N(t) - \beta t\Big).$$

For every $\beta > 0$,

$$S(\beta) \leq \mathcal{F}(\beta) \leq S(\beta) + \log(1 + \log n).$$

Proof. Fix $t > 0$. The $N(t)$ tokens with $\Delta_j \leq t$ each contribute at least $e^{-\beta t}$, thus

$$N(t)e^{-\beta t} \leq Z(\beta),$$

and hence $\log N(t) - \beta t \leq \mathcal{F}(\beta)$. After taking supremums, this gives $S(\beta) \leq \mathcal{F}(\beta)$.

Consider the ordered gaps $0 = \Delta_1 \leq \Delta_2 \leq ... \leq \Delta_n$. Since $j \leq N(\Delta_j)$,

$$\log j - \beta\Delta_j \leq \log N(\Delta_j) - \beta\Delta_j \leq S(\beta),$$

by definition of S. Thus,

$$e^{-\beta\Delta_j} \leq \frac{S(\beta)}{j},$$

which when summed over j gives

$$Z(\beta) \leq e^{S(\beta)} \leq S(\beta)\sum_{j=1}^{n}\frac{1}{j} \leq e^{S(\beta)}(1 + \log n),$$

by a classic bound on the harmonic numbers. Taking logarithms on both sides of the inequality yields the desired upper bound for $\mathcal{F}$. □

Corollary 6.3.5: If $\beta_n/\Lambda_n \to 0$ and the *contact-count entropy* $C_n = \log N\Big(\Delta_\Lambda^{(n)}\Big) = \Lambda_n\Delta_\Lambda^{(n)}$ satisfies $C_n \to \infty$, then $\mathcal{G}(\beta_n) \to 0$.

Proof. Set $r_n = \beta_n/\Lambda_n$. By the previous proposition,

$$Z(\beta_n) \geq N(\Delta_n)e^{-\beta_n\Delta_\Lambda^{(n)}} = e^{C_n - r_n C_n} = e^{(1-r_n)C_n},$$

and thus $\mathcal{F}(\beta_n) \geq (1 - r_n)C_n$. Since $r_n \to 0$ and $C_n \to \infty$, we have that $\mathcal{F} \to \infty$, which is equivalent to $\mathcal{G}(\beta_n) \to 0$ by Proposition 6.2.3. □

7 Personal Work

This section contains several exploratory results and experiments related to attention scaling. Given the breadth and difficulty of the subject, these investigations are necessarily limited in scope, but they helped develop intuition and identify questions worthy of further study.

7.1 Results on Scores

The following are various deterministic settings for which the theory of [HK26] is applied. These may not be realistic when compared to actual transformer models, but serve as a baseline for comparison. Furthermore, the realizability of these simple examples as attention scores serve as a counterexample to the belief that the class $\beta_n \asymp (\log n)^{\xi}$ is universal to the critical scaling of attention.

7.1.1 A Counterexample to Universal Poly-Logarithmic Scaling

Theorem 7.1.1: There exists a sequence of scores $\left(s^{(n)}\right)$ with $s^{(n)} \in \mathbb{R}^n$ realizable as a row of an attention score matrix $S = \frac{QK^T}{\sqrt{d_k}}$ such that the upper tail accumulation scale Λ_n exhibits super-polylogarithmic growth:

$$\frac{(\log n)^{\xi}}{\Lambda_n} \to 0, \quad \text{for all } \xi > 0,$$

and any scaling β_n avoiding rank collapse must grow faster than any polylogarithmic function of n. Consequently, the commonly observed polylogarithmic scaling class $\beta_n = (\log n)^{\xi}$ is not universal to all possible attention scores.

Proof. Consider the scores given by

$$s_j^{(n)} = -\frac{\log(j)}{n}, \quad j = 1, ..., n.$$

This gives corresponding gaps and gap-counting function

$$\Delta_j^{(n)} = \frac{\log(j)}{n}, \quad N_n(t) = \min\left(\lfloor e^{nt} \rfloor, n\right).$$

For any index j, $N_n(t)$ is constant on the interval $\left[\Delta_j^{(n)}, \Delta_{j+1}^{(n)}\right)$ and thus the maximum of $\frac{\log N_n(t)}{t}$ over this interval must occur at $\Delta_j^{(n)}$, since this quantity is strictly decreasing. The supremum therefore occurs at some index. This yields upper tail accumulation scale

$$\begin{aligned}\Lambda_n &= \sup_{t>0} \frac{\log(N_n(t))}{t} \\ &= \max_{1\le j\le n} \frac{\min\left(\log\left(e^{n\Delta_j}\right), \log(n)\right)}{\Delta_j} \\ &= \max_{1\le j\le n} n \cdot \frac{\min(\log(j), \log(n))}{\log(j)} \\ &= n.\end{aligned}$$

Clearly, for all $\xi > 0$, we have

$$\frac{(\log n)^\xi}{\Lambda_n} = \frac{(\log n)^\xi}{n} \to 0.$$

Additionally, letting $\beta_n = \left((\log n)^\xi\right)$, the free energy

$$\begin{aligned}\mathcal{F}(\beta_n) &= \log\left(\sum_{j=1}^{n} e^{-\beta_n \Delta_j}\right) \\ &= \log\left(\sum_{j=1}^{n} e^{-(\log n)^\xi \frac{\log j}{n}}\right) \\ &\ge \log\left(n e^{-(\log n)^\xi \frac{\log n}{n}}\right) \\ &= \log(n) - \frac{(\log n)^{\xi+1}}{n} \\ &\to \infty.\end{aligned}$$

By Proposition 6.2.3, this scaling β_n yields rank collapse. For realizability, the queries $q_i = \sqrt{d_k} e_1$ and keys $k_j = -\frac{\log(j)}{n} e_1$ trivially produce scores

$$s_{ij} = \frac{q^T k_j}{d_k} = -\frac{\log(j)}{n},$$

and hence are realizable as a row of an attention score matrix. □

7.1.2 Simplified Gap Structures

We now consider scores $s_j^{(n)} = g(j)$, where g does not depend on n. In this case, all of the first n components of the score vectors $s_j^{(n)}$ and $s_j^{(n+1)}$ are identical. We are interested in the behaviour of Λ_n with respect to g.

Proposition 7.1.2: Let g be an increasing function independent of n such that[3] $g(1) = 0$ and consider scores $s_j^{(n)} = -g(j)$, yielding gaps $\Delta_j^{(n)} = g(j)$. Define for $j \geq 1$

$$\lambda_j = \frac{\log(j)}{g(j)},$$

with convention that $\lambda_1 = 0$. Then, the following hold:

1. If λ_j attains a finite maximum at index j_0, then for large enough n, the upper tail accumulation scale eventually satisfies $\Lambda_n = \lambda_{j_0} = O(1)$.
2. If $\lambda_j \to C \in [0, \infty)$ as $j \to \infty$, then $\Lambda_n = O(1)$.
3. If λ_j is increasing and $\lambda_j \to \infty$ and $j \to \infty$, then $\Lambda_n = \lambda_n$.

Proof. First, observe that for any integer k, $N_n(t) = k$ if and only if $g(k) \leq t < g(k+1)$. As in the proof of Theorem 7.1.1, $N_n(t)$ is constant on intervals $[g(k), g(k+1))$, and thus Λ_n is necessarily attained at some left endpoint $g(k)$. Thus,

$$\Lambda_n = \max_{1 \leq k \leq n} \frac{\log(k)}{g(k)} = \max_{1 \leq k \leq n} \lambda_k.$$

1. Since λ_j attains a finite maximum, then necessarily $\Lambda_n = \lambda_{j_0}$ eventually.
2. Because $\lambda_j \to C$ as $j \to \infty$, Λ_n is bounded by a basic property of sequences.
3. Since λ_j is increasing, necessarily $\Lambda_n = \lambda_n$ for all n.

□

This proposition identifies $g(j) = \log(j)$ as the boundary between bounded and divergent upper-tail accumulation scales Λ_n. Indeed, if $g(j) = \log(j)$, then $\lambda_j = 1$, so $\Lambda_n = 1$. However, if $g(j) = (\log j)^{1-\varepsilon}$ then $\lambda_n = (\log n)^{\varepsilon} \to \infty$ and hence $\Lambda_n = (\log n)^{\varepsilon}$. Note that this is not related to any discussion of the proposed attention scaling $\beta_n \asymp \log n$.

Furthermore, in this context, we observe that the gap counting function is given by

$$N(t) = \#\{j : g(j) \leq t\} = \#\{j : j \leq g^{-1}(t)\} = \min(\lfloor g^{-1}(t) \rfloor, n),$$

and hence the growth of this function depends on the inverse of g. For sub-logarithmic functions, the inverse, and hence $N(t)$ grows super-exponentially and, causing collapse in weights after softmax. In this case, scaling is necessary. For super-logarithmic functions, the inverse grows sub-exponentially and $N(t)$ grows at an appropriate weight for softmax to avoid collapse.

[3]Recall that this assumption can be made because of the shift invariance of softmax.

Proposition 7.1.3: Let g be an increasing function independent of n such that $g(1) = 0$ and a_n a positive sequence. Define scores

$$s_j^{(n)} = -a_n g(j), \quad r_j^{(n)} = -g(j)$$

and let $\Lambda_n^{(s)}$ and $\Lambda_n^{(r)}$ denote their corresponding upper tail accumulation scales. Then,

$$\Lambda_n^{(s)} = \frac{1}{a_n} \Lambda_n^{(r)}.$$

More generally, if $r^{(n)}$ is any score sequence with $\Lambda_n^{(r)} < \infty$ and $s^{(n)} = -a_n r^{(n)}$, then the above holds.

Proof. Observe the relation

$$N^{(s)}(t) = \#\{j : a_n g(j) \leq t\} = \#\{j : g(j) \leq t/a_n\} = N^{(r)}(t/a_n).$$

By a change of variables $u = t/a_n$,

$$\Lambda_n^{(s)} = \sup_{t>0} \frac{\log\big(N^{(s)}(t)\big)}{t} = \sup_{t>0} \frac{\log\big(N^{(r)}(t/a_n)\big)}{t} = \frac{1}{a_n} \sup_{u>0} \frac{\log\big(N^{(r)}(u)\big)}{u} = \frac{1}{a_n} \Lambda_n^{(r)}.$$

□

This proposition is of practical importance in the following setting. Consider scores $s^{(n)}$ whose components can be decomposed into a index dependent part $g(j)$ and a global scaling part a_n such that $s_j^{(n)} = -a_n g(j)$. Then, this proposition applies and we immediately obtain Λ_n by first applying the sequence criterion, then scaling by $1/a_n$.

Corollary 7.1.4: Let b_n be a positive sequence only depending on n. Then, the scores

$$s_j^{(n)} = -\frac{1}{b_n} \log(j).$$

have $\Lambda_n = b_n$ and the critical scaling satisfies $\beta_n \asymp b_n$.

Proof. By Proposition 7.1.2, the scores $r_j^{(n)} = -\log(j)$ have $\Lambda_n = 1$ since $\lambda_j = 1$ for all j. By the previous proposition, the upper tail accumulation scale for $s_j^{(n)}$ is $\Lambda_n = b_n$. The fact that $\beta_n \asymp b_n$ is the result of Theorem 6.1.4. □

Although potentially unrealistic in the context of real models, this shows that is possible to construct a sequence of score vectors with any desired critical scaling.

Corollary 7.1.5: Let $L_n, f(n) > 0$. If the unscaled gap sequence satisfies $\Lambda_n^{(r)} = \Theta(L_n)$, then the rescaled scores $s^{(n)} = a_n r^{(n)}$ with $a_n = L_n/f(n)$ satisfy

$$\Lambda_n^{(s)} = \Theta(f(n)).$$

Proof. An immediate consequence of Proposition 7.1.3. □

7.1.3 Applied Examples

7.1.3.1 Linear Gaps (Unbounded)

Consider scores

$$s_j^{(n)} = -(j-1),$$

which is to say $g(j) = j - 1$ in the context of the theory above. Then,

$$\Lambda_n = \max_{1 \le k \le n} \frac{\log(k)}{k-1} = \log 2.$$

7.1.3.2 Linear Gaps (Bounded)

Consider scores

$$s_j^{(n)} = -\frac{j-1}{n}.$$

This is the last example scaled by $\alpha_n = \frac{1}{n}$, meaning

$$\Lambda_n = n \log 2,$$

by Proposition 7.1.3. We may also analyze this setting with respect to the maximum weight

$$a_{\max} = \frac{1}{Z(\beta_n)} = \left[\sum_{j=1}^{n} \exp\left(-\beta_n \frac{j-1}{n}\right)\right]^{-1} = \frac{1 - e^{-\beta_n/n}}{1 - e^{-\beta_n}},$$

where if $\beta_n \to \infty$ and $\beta_n/n \to \lambda$ as $n \to \infty$,

$$a_{\max} \to \begin{cases} 0, & \lambda = 0, \\ 1 - e^{-\lambda}, & \lambda \in (0, \infty), \\ 1, & \lambda = \infty. \end{cases}$$

In particular, the scaling $\beta_n = n \log 2$ will have maximum weight $a_{\max} = \frac{1}{2}$.

7.1.4 Super-Logarithmic Gaps

By Proposition 7.1.2, any function such that

$$\frac{\log(j)}{g(j)} \to 0$$

as $j \to \infty$ will necessarily have $\Lambda_n = O(1)$, and rescalings by α_n will simply affect Λ_n by a factor of $1/\alpha_n$. This covers a wide range of functions (exponential, linear, polynomial, etc.).

7.2 Connection to Statistical Physics and Simulated Annealing

7.2.1 Setup

The softmax operator admits a natural interpretation through the framework of statistical physics. In this context, a system with states x and and energy function $H(x)$ is described by the Gibbs distribution

$$G_{T(x)} = \frac{e^{-H(x)/T}}{Z(T)},$$

where T is the temperature and $Z(T)$ is known as the partition function

$$Z(T) = \sum_x e^{-H(x)/T}.$$

Proposition 7.2.1: Given a finite set of states x with energies $H(x)$, the Gibbs distribution is the distribution that maximizes the entropy

$$S(x) = -\sum_x p(x) \log p(x)$$

among all probability distributions p satisfying a fixed expected energy constraint

$$\sum_x p(x)H(x) = U.$$

Proof. We use the method of Lagrange multipliers for the two constraints. Define

$$\mathcal{L}(p, \alpha, \beta) = -\sum_x p(x) \log p(x) - \alpha\left(\sum_x p(x) - 1\right) - \beta\left(\sum_x p(x)H(x) - U\right).$$

For each x,

$$\frac{\partial \mathcal{L}}{\partial p(x)} = -(\log p(x) + 1) - \alpha - \beta H(x),$$

which when set to zero gives

$$p(x) = Ce^{-\beta H(x)}.$$

Under the normalization condition, $\sum_x p(x) = 1$, it is clear that

$$C = \frac{1}{\sum_x e^{-\beta H(x)}}.$$

Finally, we note that

$$\frac{\partial^2 S}{\partial p(x)^2} = -\frac{1}{p(x)} < 0,$$

and

$$\frac{\partial^2 S}{\partial p(x)\partial p(y)} = -\frac{1}{p(x)} = 0,$$

for $y \neq x$, hence the Hessian of S is negative definite and S is strictly concave. The two constraints are linear, so the feasible set is convex. Thus, this distribution is indeed the minimum. The multiplier β can be reparameterized by setting $\beta = 1/T$, as is common in physics, and is the constraint that controls the average energy of the distribution. □

The attention distribution has exactly this form when distances to the maximum score are treated as energy levels,

$$a_j = \frac{e^{-\beta\Delta_j}}{\sum_{k=1}^{n} e^{-\beta\Delta_k}},$$

as seen before. In this context, attention acts as a physical system in which both entropy and energy are key quantities. This is a fundamental theme in the domain of physics and provides an explanation for the ideas of attention scaling.

First, the gap counting function $N(t)$ gives the amount of scores (states) available below a certain threshold (energy level). Restricted to this energy level, the maximum entropy distribution over these states is that of the uniform distribution,

$$S(t) = \log N(t).$$

Using the above interpretation,

$$\Lambda_n = \sup_{t>0} \frac{S(t)}{t}$$

is the largest entropy increase per unit energy tolerance. Second, the scaling β controls the average energy of the output distribution, as seen above. Increasing β strengthens the preference for low-energy states, since higher-energy states are exponentially suppressed by the factor $\exp(-\beta H(x))$. For a fixed n, the zero-temperature limit $\beta \to \infty$ leads to a distribution that concentrates on only the minimal energy states, which is to say the poentially tied maximal attention scores, and disregards all other scores.

At distance t from the maximum, the energy penalty is $e^{-\beta t}$, and the number of choices gained is $N(t)$. The total contribution of this set to the partition function $Z(\beta)$ is therefore bounded below by $N(t)e^{-\beta t}$, which exhibits two competing exponential effects, the growth in number of available near-optimal states and the energetic penalty for moving away from the optimum. Taking logarithms gives the effective competition

$$\log N(t) - \beta(t) = S(t) - \beta t = t\left(\frac{S(t)}{t} - \beta\right).$$

Thus, the critical scaling occurs when these two effects balance. If $\beta_n \gg \Lambda_n$, then $\frac{S(t)}{t} - \beta < 0$ for all t, meaning the energy penalty dominates the entropy gained from additional states. The distribution concentrates on the lowest-energy states, leading to attention collapse.

Conversely, if $\beta_n \ll \Lambda_n$, there is energy levels t for which $\frac{S(t)}{t} - \beta > 0$. At such a level, the exponential growth in the number of accessible states outweighs the exponential suppression of their individual weights. Consequently, collections of near-optimal states make a non-neglible contribution to the partition function, preventing minimal-energy states from being the unique dominant contributors. In particular, if we are considering a sequence of scores for which $\Lambda_n \to \infty$, $\beta_n \ll \Lambda_n$ implies that

7.2.2 Simulated Annealing

Simulated annealing is a stochastic optimization method motivated by the physical process of thermal annealing, in which a material is gradually cooled to reach a low-energy equilibrium configuration. The method considers an optimization problem defined by an *energy* or *cost* function $E(x)$, where x denotes a possible configuration and lower values of $E(x)$ correspond to more desirable solutions. The objective is to identify a global minimizer $x^* = \arg\min E(x)$.

Unlike deterministic optimization methods that always select an improving transition, simulated annealing allows occasional transitions to higher-energy configurations. Given a current state x, a candidate state x' is generated according to a problem-dependent proposal distribution $x' \sim q(x'|x)$. The proposed transition is accepted with probability

$$\mathbb{P}(x \to x') = \begin{cases} 1, & E(x') \leq E(x) \\ \exp\left(-\frac{E(x')-E(x)}{T}\right), & E(x') > E(x) \end{cases}$$

where T is a parameter called *temperature*. At high temperatures, the algorithm frequently accepts higher-energy states, allowing for broad exploration of state space and escaping from local minima. As temperature decreases, higher-energy transitions become increasingly unlikely, causing the system to concentrate on low-energy configurations.

It can be shown [CITATION NEEDED] that the Gibbs distribution,

$$p(x) = \frac{e^{-E(x)/T}}{\sum_x e^{-E(x)/T}}$$

is the stationary distribution of the Markov chain. In other words, for a fixed T and under standard assumptions on the transition mechanism, repeated application of the transition rule causes the distribution of visited states to converge to p. Thus, simulated annealing can be interpreted as a procedure for sampling from the Gibbs distribution at a given temperature.

The defining feature of simulated annealing is the gradual reduction of the temperature parameter t. The temperature is therefore not held fixed, but is reduced according to a *cooling schedule* $T_0 > T_1 > T_2 > \ldots > T_K$. At each temperature T_k, the Markov chain is allowed to approach equilibrium before the temperature is lowered further. This creates a sequence of Gibbs distributions $p_1, \ldots, p_K$ that gradually shifts probability mass from exploration of the entire state space towards exploration of the lowest-energy configurations.

Under certain assumptions, sufficiently slow logarithmic cooling guarantees convergence to a global minimum. However, this schedule is often impractical because it requires an extremely large number of iterations. In practice, faster schedules such as geometric cooling $T_{k+1} = \alpha T_k$ are typically used because they provide a better tradeoff between computational cost and solution quality. Further research showed that the cooling schedule largely depends on the geometry of the cost function.

In terms of attention, we may interpret the indices $j = 1, ..., n$ as a set of possible states, with the energy of each state defined by its score gap $E(j) = \Delta_j$. The softmax operation then induces a Gibbs distribution

$$p(j) = a_j = \frac{e^{-\beta \Delta_j}}{\sum_{j=1}^{n} e^{-\beta \Delta_j}},$$

where $\beta = 1/T$ acts as an inverse temperature.

However, there is an important distinction from simulated annealing. Simulated annealing is an optimization procedure that gradually modifies the temperature of a Markov chain in order to concentrate probability mass around global minima of an energy function. Attention does not perform an iterative search, nor does it evolve a state distribution through a Markov process. Instead, the distribution is computed in a single forward pass from the learned query-key interactions. Therefore, the temperature analogy should not be interpreted as an optimization mechanism, but rather as a characterization of how the learned attention landscape controls the selectivity of a given attention head.

The question is therefore not whether attention directly implements or should implement simulated annealing, but whether the learned model parameters implicitly exhibit or benefit from a (not necessarily monotonic) temperature schedule that varies with either layer depth l or context length n.

7.2.3 Role of Constants in Λ_n

8 Experiments on Scores

The experiments in this section investigate the empirical behaviour of the theoretical quantities introduced in previous sections. We evaluate these quantities across four pretrained Transformer models: GPT-2 Medium and GPT-2 Large [Rad+19], OPT-350M [Zha+22], and BERT-Large [Dev+19]. The relevant architectural characteristics of these models are summarized in [REF TABLE]. The scope of the analysis was limited to these models due to the substantial computational requirements of attention-based analysis for larger architectures.

Model	**Architecture**	**Parameters**	**Layers**	**Heads**	**Embed Dim.**
GPT-2 Medium	Causal	355M	24	16	1024
GPT-2 Large	Causal	774M	36	20	1280
OPT-350M	Causal	331M	24	16	1024

Model	Architecture	Parameters	Layers	Heads	Embed Dim.
BERT-Large	Bidirectional	340M	24	16	1024

This selection allows for the following comparisons:

- Scale: GPT-2 Medium and GPT-2 Large share the same architecture while differing primarily in model size and depth.
- Implementation and Training: GPT-2 Medium and OPT-350M have comparable architectural dimensions, but were trained using different datasets and training procedures.
- Masking: GPT-2 models and OPT-350M use causal self-attention (triangular masking), where each token attends only to previous tokens. BERT-Large uses bidirectional self-attention (no masking), allowing each token to attend to the full input sequence.

The input sequences used for evaluation are summarized in [REF TABLE]. The selected sequences are designed to probe different attention behaviours.

Due to the computational cost of extracting and analyzing full attention distributions across multiple models, the evaluation is restricted to relatively short input sequences. Longer contexts may reveal more behaviour, but the selected sequences are likely sufficient to investigate the layer-wise structure of relevant quantities at finite scales (i.e. not in the $n \to \infty$ limit).

ID	Type	Purpose	Example excerpt
P1	Natural language	Baseline text	“The scientist walked into the laboratory…”
P2	Narrative	Long-range dependencies	“In the summer of 1847, a young researcher began…”
P3	Repetition	Pattern matching	“red blue green yellow red blue…”
P4	Retrieval	Context recall	“Alice went to the market. She bought apples…”
P5	Mathematical reasoning	Symbolic reasoning	“If x equals 5 and y equals 10…”
P6	Code	Syntax structure	“def factorial(n): if n == 0…”
P7	Factual text	Semantic integration	“The history of mathematics is a story…”
P8	High entropy	Incoherent text	“purple quantum bicycle sings silently…”
P9	Random tokens	Unpredictable input	“xylophone nebula 47 carpet velocity…”
P10	Logical reasoning	Abstract relations	“Every A is a B. Every B is a C…”

We will use the notation $\Theta_{p,l,h,q}$ to refer to some observable quantity Θ for prompt p, layer l, attention head h, and token query q. Models will be analyzed separately then compared unless otherwise stated.

8.1 Effective Scaling

Recall the learned parameters W_Q and W_K in the attention score calculation

$$S = \frac{1}{\sqrt{d_k}} X W_Q (X W_K)^T.$$

Assuming classical softmax with no explicit scaling, the model may implicitly learn a scaling β through the norms of the W_Q and W_K matrices, or potentially through earlier transformations such as the input embedding matrix. For example, scaling either matrix by a constant scales the attention scores by the same factor. Since the entire model is a parameterized system, this scaling is not necessarily represented by a single identifiable paramter in a pre-trained model. Even if the parameter β is included in the model and trained, the effective scaling of attention scores may still be controlled by earlier parameters if no explicit regularization is included. Furthermore, this is not limited to being a scalar, since different rows of the attention matrix differently by modifying the weights of the matrices. Thus, extracting a true underlying temperature for a fixed context length n for a given attention head requires defining a reference frame and some compromise in aggregation.

Up to shifting, softmax can be inverted to obtain scores $s_j = \log(a_j)$

Definition 8.1.1: Given attention scores $s \in \mathbb{R}^n$, define the effective scaling

$$\beta_{\text{eff}} = \Lambda^{-1}$$

where Λ is the corresponding upper tail accumulation scale of s. For some aggregated scale $\overline{\Lambda}$ over many individual scales Λ, the corresponding effective scaling is $\overline{\Lambda}^{-1}$.

Assuming that the model keeps $\Lambda^{(r)} = 1$ in an unscaled reference state with scores r_j, and has scaled scores $\beta_{\text{eff}} \cdot s_j$ according to some effective scaling, applying Proposition 7.1.3 gives that $\beta_{\text{eff}} = 1/\Lambda^{(s)}$. This can also be interpreted as the ratio $r = \beta_n / \Lambda$ with the actual value $\beta = 1$, which has $r \ll 1$ and $r \gg 1$ corresponding to "hot" high-entropy and "cool" low-energy states, respectively, and is the ratio investigated asymptotically in Theorem 6.1.4. Other methods to define an effective scaling are possible, such as the standard deviation $\sqrt{\operatorname{Var}(s)}$ or the matrix norms $\|W_q\|$ or $\|W_k\|$, however these do not seem to appropriately capture the scaling effects.

Noting that softmax can be inverted up to shifting to obtain gaps $\Delta_j = -\log(a_j)$, this metric allows Λ acts as an observable quantity that gives information on scaling beyond the theoretical quantification of collapse. This serves an important tool in interpreting the models temperature scheduling. We will also use $T_{\text{eff}} = \beta_{\text{eff}}^{-1} = \Lambda$ in order to draw parallels with simulated annealing.

8.2 Upper Tail Accumulation Scale

The question is naturally how $\Lambda_{p,l,h,q}$ varies

8.2.1 Lambda Cooling

We aggregate over prompts, layers, and queries to obtain the layer average

$$\Lambda_l = \frac{1}{phq} \sum_{p,h,q} \Lambda_{p,l,h,q}.$$

While this is suceptible to many oversmoothing because of averaging, it is a starting point in investigating this quantity.

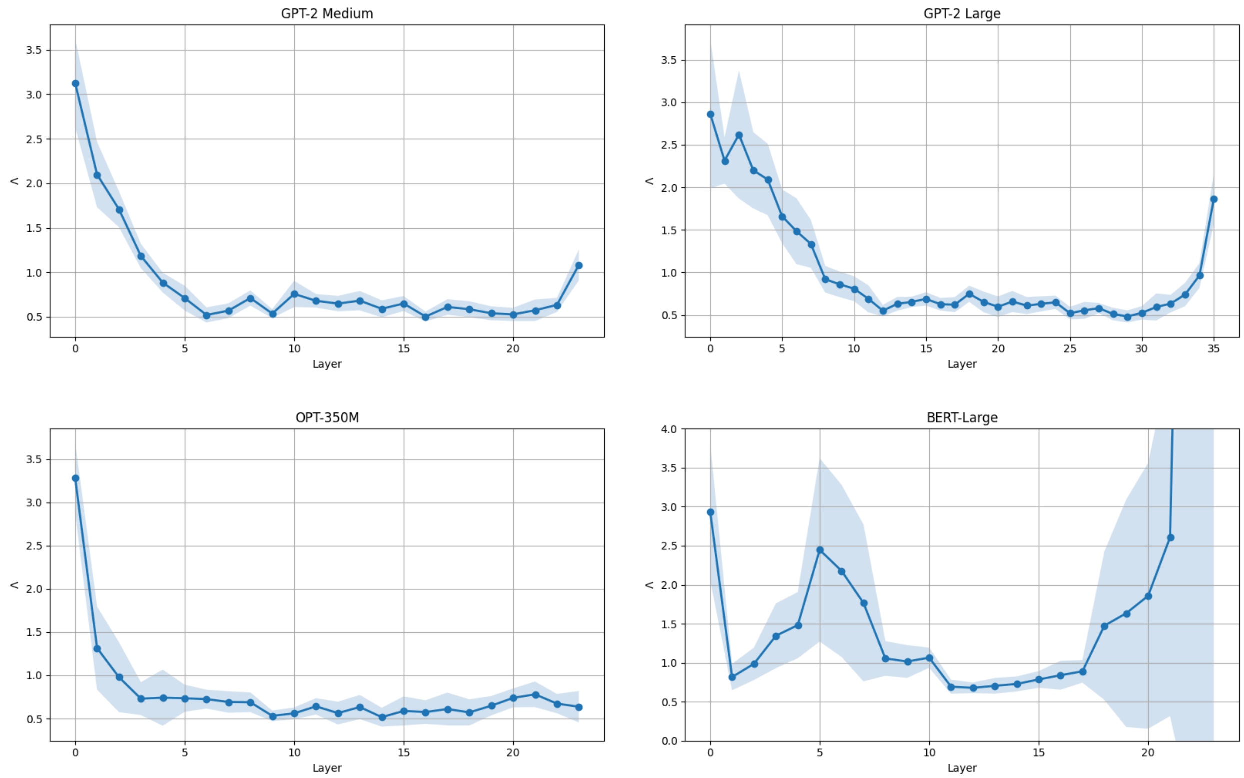

Figure 8: Evolution of mean Λ by layer. Shaded regions show ± 1 std across prompts.

Interpreting Λ as T_{eff}, the three causal (masked) models qualitatively appear to apply a cooling process as depth increases. As a first observation, this suggests that this initial cooling phase is an important part of effective text-based causal transformer models. Theoretically, this means that if a model were to naively scale scores by their corresponding scales Λ^{-1} before applying standard softmax, in turn leading to scaled scores having $\Lambda = 1$, then the effective cooling schedule would be constant and it is possible the model would underperform.

The two GPT models have roughly corresponding trajectories in terms of relative length of the apparent phases. Aside from the initial cooling, a particularly interesting phase is the "reheating" phase in the final few layers of the models. This is noticeably different to the final few layers in OPT-350M. The GPT models and OPT-350M are largely similar in architecture, but were trained on substantially different datasets, which may explain this difference.

The more erratic results of the bidirectional (unmasked) BERT model indicate that behaviour is more complex than expected. For this reason, the following subsections will be restricted to the causal models and the corresponding experiments on BERT will be discussed in a later subsection.

8.2.2 Input Prompt Dependence

As shown in Figure 8, the three causal models have relatively small deviation from the mean across the 10 prompts. This suggests that the layer-wise evolution of Lambda is primarily a property of model architecture and depth rather than the specific input sequence. Figure 9 plots the averaged $\Lambda_{p,l}$ and shows that all prompts follow a roughly similar trajectory. The plots for the two other causal models are similar and have not been included. Minor deviations in overall vertical shift are explained by sequence length n, and per-layer deviations such as that of prompt 3 in layer 10 could be explained by how the specific prompt (in this case, a repetitive sequence) interacts with a layer's learned dynamics. These deviations are neglible compared to the overwhelming trend of following the mean trajectory, but might be an interesting direction for future work.

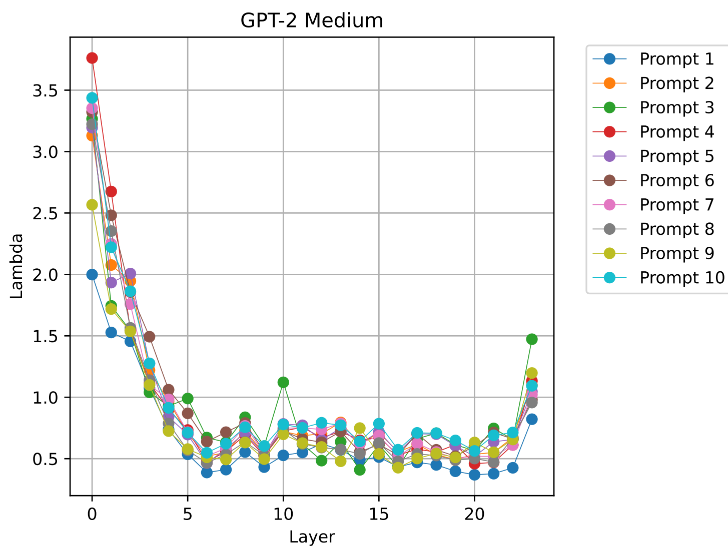

Figure 9: Mean Λ by layer and prompt.

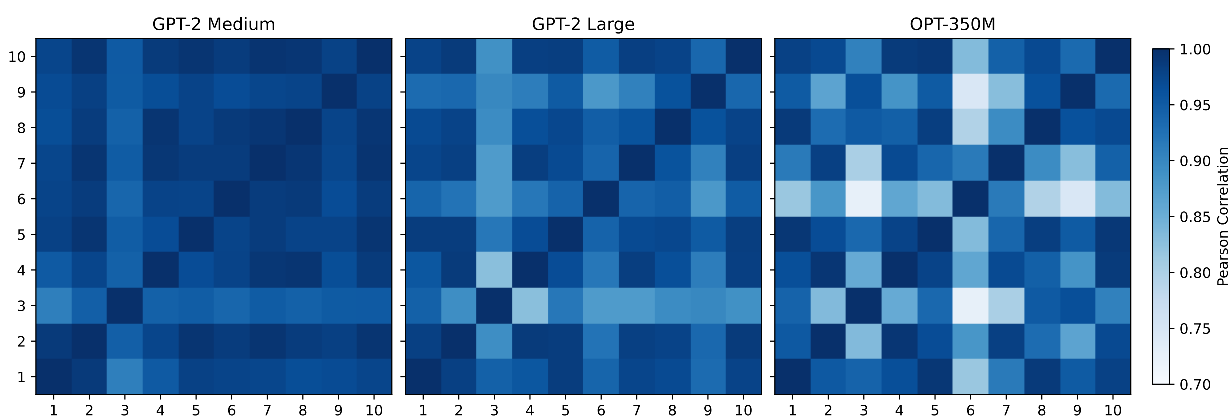

Figure 10: TODO

8.3 Longer Context

8.4 Comparison with Other Metrics

For consistency, it is important to verify that this measure of effective scaling correlates to the entropy of the attention weights.

8.4.1 Longer Sequences

8.4.2 BERT

9 Discussion

9.1 Alternatives to Softmax

- Top x: issue doesn't scale well
- Sparsemax?

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